HERMITIAN–YANG–MILLS FLOW AND GLUING ON DEGENERATING CURVES

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Abstract. For a six-parameter degeneration of plane quartics, we approximate Hermitian–Yang–Mills metric flow on flat unitary bundles with trivial component connections and noncommuting gluing by a finite gradient flow. Its energy is a sum of squared distances between positive Hermitian matrices after unitary transport, weighted by neck capacities. The integrated squared-distance comparison holds on time intervals proportional to the shortest logarithmic neck length, uniformly in all node-length ratios and as the gluing representation becomes reducible. We prove relative asymptotics for every small linearized eigenvalue and reach the comparison from an open class of bounded spatial initial metrics with uniform component regularity after a negligible initial layer. The comparison holds for all time when the smallest effective eigenvalue times the shortest logarithmic neck length stays bounded away from zero. A second compact family permits independently vanishing component holonomy and neck size. On the quartic family with equal node parameters we determine the leading Hodge metric on the unitary Schottky locus and asymptotically recover projective gluing from low eigenvalues and component averages of orthonormal eigensections in fixed unitary frames. On a finite class of first-order rational-component smoothings, these observations and the base-loop orientation determine unknown adjacency and the integral symplectic conjugacy class of monodromy.

1. Introduction

When a compact Riemann surface develops nodes, a metric on a bundle can equilibrate on the normalization components long before the components equilibrate with one another. The remaining variables are matrix metrics. Their coupling depends on the fibre identifications at the nodes, including cycle holonomy that cannot be removed by component frame changes. We prove a finite nonlinear reduction of the actual Hermitian–Yang–Mills metric flow in this setting. The comparison allows independent node lengths and noncommuting unitary gluing. It is accompanied by a complete relative spectral theorem, entry from spatial initial metrics, and geometric and inverse consequences.

The principal family smooths four projective lines in general position. Its six nodes have independent exact holomorphic equations $u_e v_e = \tau_e$; its smooth fibres C_τ are plane quartics of genus three. We use the metric induced by the Fubini–Study form, normalized so that a line has area one. The construction in Section 2 gives a proper projective family and, for every tuple $U = (U_e) \in \mathrm{SU}(r)^6$, a holomorphic bundle with flat unitary connection D_U, parallel metric h_0 and trivialized determinant. Here $r \geq 2$ is fixed. The component connections are trivial, and fibre holonomy factors through the free group of rank three of the dual graph K_4.

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The bundles are polystable of degree zero and are stable exactly when this representation is irreducible.

1.1. **The effective equation and the main theorem.** Transport U_{ij} takes the j component frame to the i frame. Set

$$L_e = \log(1/|\tau_e|), \qquad L_* = \min_e L_e, \qquad w_e = 2\pi/L_e.$$

All parameters are fixed during each flow, and the degeneration limit is $L_* \to \infty$. Let A_i denote the limiting component areas, equal to one for the chosen Fubini–Study metric. On the real space of trace-free Hermitian tuples $\mathcal{V} = \bigoplus_{i=0}^{3} \mathrm{Herm}_0(r)$ define

$$M(X) = \sum_i A_i \,\mathrm{tr}(X_i^2), \qquad Q_{\tau,U}(X) = \sum_{e=ij} w_e \|X_i - U_{ij} X_j U_{ij}^*\|_{\mathrm{HS}}^2.$$

Let $\nu_0 \le \cdots \le \nu_{m-1}$ be the generalized eigenvalues of $(Q_{\tau,U}, M)$, including zeros, where $m = 4(r^2-1)$. The actual linearized operator is $\Delta_{\tau,U} = D_U^* D_U$ on trace-free h_0-Hermitian endomorphisms, with the unnormalized area norm. Its eigenvalues are denoted by λ_k.

Write $\mathcal{P}_r$ for the determinant-one positive Hermitian matrices, with symmetric-space metric $g_P(V,W) = \mathrm{tr}(P^{-1}VP^{-1}W)$ and distance d. The finite potential and its negative gradient equation, using the area-weighted product metric, are

$$\begin{aligned} \Phi_{\tau,U}(\mathbf{H}) &= \frac{1}{2} \sum_{e=ij} w_e d(H_i, U_{ij} H_j U_{ij}^*)^2, \\ H_i^{-1} \dot{H}_i &= \frac{1}{A_i} \sum_{e=ij} w_e \log(H_i^{-1} U_{ij} H_j U_{ij}^*). \end{aligned} \tag{1}$$

The exact metric flow uses physical time s and the normalization $H^{-1}\partial_s H = -2i\Lambda F_H$. For two metrics set $\sigma(H,K) = \mathrm{tr}(H^{-1}K + K^{-1}H) - 2r$. It is intrinsic under frame changes and bounds $d(H,K)^2$ from above.

We also specify an initial class that does not require an already matched finite tuple. Put $\rho = L_*^{-1}$, fix $0 < \beta < 1$ and $R_0 > 0$, and write an initial metric as $h_0 e^S$. Let $\Omega_i(2\rho) \subset C_i$ be the complement of the incidence discs of radius 2ρ. Pull S back to these component exteriors using the fixed geometric identifications and flat frames of Section 2. Require

$$\|S\|_{C^0(C_\tau)} + \inf_{\widetilde{S}_i|_{\Omega_i(2\rho)} = S_i} \sum_i \|\widetilde{S}_i\|_{C^{6,\beta}(C_i)} < R_0, \tag{2}$$

where the extensions are trace-free Hermitian matrices on the fixed components. This is a nonempty open class on each smooth fibre. Its radius is independent of the degeneration; bounded smooth data on the middle necks may have arbitrarily large derivatives.

A unitary frame identifies each endomorphism fibre with $\mathrm{Mat}_r(\mathbb{C})$ and so specifies the matrix-algebra marking used below. Projective transport means the class of a unitary edge map in $\mathrm{PSU}(r) = \mathrm{SU}(r)/Z(\mathrm{SU}(r))$. After choosing a base vertex, cycle transports are represented in its frame; changing that frame conjugates all cycle maps simultaneously.

Theorem 1.1. *The compact quartic family and flat bundles above have the following properties.*

(i) *There are $\gamma > 0$ and $\epsilon(\tau) \to 0$ as $L_* \to \infty$ such that*

$$(1 - \epsilon(\tau))\nu_k \le \lambda_k \le (1 + \epsilon(\tau))\nu_k \quad (0 \le k < m), \qquad \lambda_m \ge \gamma.$$

The estimates are uniform in all node-length ratios and all U. The kernel is naturally identified with the space of tuples satisfying $X_i = \mathrm{Ad}(U_{ij}) X_j$.

(ii) *For finite initial tuples in a fixed bounded ball,* (1) *has a confined all-time solution. It determines a smooth determinant-one comparison metric* $\widehat{K}(s)$ *on the entire* C_τ*, converging to* $H_i(s)$ *on fixed component compact subsets with error* $O(L_*^{-1})$*. If the exact initial metric has fixed positive bounds against* h_0 *and* $\eta = \int_{C_\tau} \sigma(H^{\mathrm{in}}, \widehat{K}(0))\, dA_\tau$*, then, with* $e_* = (1+\log L_*)^2/L_*$*,*

$$\sup_{0\le s\le TL_*} \int_{C_\tau} d(H(s), \widehat{K}(s))^2\, dA_\tau \le \eta + CTe_* \qquad (T \ge 0).$$

The constants are independent of T*, the node-length ratios and the unitary transports.*

(iii) *For the spatial initial class* (2)*, the component flows select a bounded tuple* P *and there is a uniform preparation time* $s_* \to \infty$*,* $s_*/L_* \to 0$*, such that the comparison in part* (ii) *applies to* $H(s_* + s)$ *and the finite flow starting at* P*, with initial error* $\eta_* \to 0$ *uniformly over that class.*

(iv) *If* $L_*\nu_0 \ge \kappa > 0$*, then for the initial data and discrepancy in part* (ii)

$$\sup_{s\ge 0} \int_{C_\tau} d(H(s), \widehat{K}(s))^2\, dA_\tau \le C\eta + C_\kappa e_*(1 + |\log e_*|).$$

The same conclusion holds after the preparation in part (iii)*, with* η_* *in place of* η*.*

(v) *On the common-node slice* $\tau_e = t$*, put* $L = \log(1/|t|)$*. The graph representation quotient* $\mathrm{SU}(r)^{6,\mathrm{irr}}/\mathrm{SU}(r)^4$ *embeds as a Lagrangian locus in the irreducible flat-bundle moduli space of* C_t*. Its pullback Hodge squared norm is* $2\pi/L$ *times the quotient of the bi-invariant product squared norm on* $\mathrm{SU}(r)^6$*, with relative error* $O(L^{-1})$ *uniform towards reducibility. Moreover, the first* m *eigenvalues and the component averages of their* L^2*-orthonormal eigensections, in fixed unitary matrix-algebra frames, consistently determine the projective edge transports. Cycle transports are recovered up to root conjugation. Central lifts are not determined by these observations.*

Proof. Propositions 2.1 and 2.3 construct the family, bundle and holonomy representation. Theorem 3.1 proves part (i). Proposition 4.1 and Theorem 5.2 prove part (ii); Theorems 6.1 and 6.2 give the component limits and preparation in part (iii). Theorem 5.3 gives part (iv). The moduli statement is Theorem 8.1, and the observed-operator construction of Lemma 9.2 together with Theorem 9.3 gives the last assertion. All these constructions use the same quartic family, flat bundles and metric normalization. □

The relative estimate in part (i) controls every positive small eigenvalue even when its ratio to the largest weight tends to zero. It does not imply nonlinear tracking on the inverse of that weakest eigenvalue. For example, Proposition 3.5 realizes orders h, h^2, h^4 on actual stable rank-two bundles. The uniform all-time hypothesis in part (iv) fails in that example. Likewise, part (v) concerns the indicated moduli locus and marked observations. It does not assert an ambient moduli completion or reconstruction from eigenvalues alone.

1.2. **Component holonomy and integral monodromy.** Two further constructions test different aspects of the reduction. In Section 7 a compact genus-three family has two elliptic and two rational normalization components. The component flat connections vary with parameters α_i independently of the common node parameter. The effective energy contains the component commutator terms of order α_i^2 and the transported edge terms of order L^{-1}. Theorems 7.2, 7.3 and 7.4 prove the complete relative spectrum, actual nonlinear comparison and spatial entry for arbitrary ratios of these scales. The nonlinear error on bounded intervals

in s/L tends to zero as $\max_i |\alpha_i| + L^{-1} \to 0$. This family has a common node scale; it does not combine varying component holonomy with all six independent quartic parameters.

The topology problem in Section 9 starts with six labelled rational components, each with three unpaired branch points. Their pairing is unknown. We construct compact first-order smoothings for a finite class of simple trivalent pairings, with identical prescribed component metrics and local node metric germs. The first $6(r^2-1)$ eigenvalues and component averages of the eigensections determine the missing adjacency for all sufficiently small parameters. The known first-order smoothing and positive base-loop orientation then determine the integral symplectic conjugacy class of monodromy on first homology.

For the triangular prism and $K_{3,3}$ the torsion groups of the cokernel of monodromy minus the identity are $\mathbb{Z}/5\mathbb{Z} \oplus \mathbb{Z}/15\mathbb{Z}$ and $(\mathbb{Z}/3\mathbb{Z})^2 \oplus \mathbb{Z}/9\mathbb{Z}$. Both smooth fibres have genus four; the distinguished invariant belongs to the degeneration, not to their diffeomorphism types. The classical Picard–Lefschetz and graph-lattice identifications [AO20, BHN97] are used after the analytic observations have supplied the missing incidence. These same compact families also carry the noncommuting HYM dynamics, by Proposition 9.1.

1.3. **Relation to earlier work and the proof.** Relative graph descriptions of small surface eigenvalues have a substantial history. Burger [Bur90, Theorem 1.1] proves relative scalar asymptotics on degenerating hyperbolic surfaces without a lower bound on the ratios of the short geodesic lengths. Thus neither a graph energy nor arbitrarily separated scalar scales is new by itself. Our surface metrics are induced from a smooth ambient metric, the coefficients act on bundle endomorphisms, and the holonomy kernel may change. Although the Dirichlet energy in two dimensions is conformally invariant, the function L^2 mass is not; the ambient-metric normalization requires its own comparison.

For the spectral theorem, component averages give the finite variables and logarithmic interpolation gives trial sections with the prescribed edge energy. The reverse comparison bounds each boundary mismatch by the energy of the actual section before summing with the individual edge weights. This avoids division by the smallest weight. A uniform annular mass estimate rules out additional weak modes; min–max then compares the entire small cluster relatively, including changing zero eigenspaces.

Daskalopoulos–Wentworth [DW00] study connection spectra and Yang–Mills flow near the boundary of Teichmüller space. The component-trivial connections considered here have a large limiting parallel space even when the global representation is irreducible. We estimate the slow coupling of that space and its nonlinear metric evolution. The fixed-base asymptotic theories of Haiden–Katzarkov–Kontsevich–Pandit [HKKP18] and Lam [Lam26] provide a complementary reduction of HYM evolution. Their fixed-background estimates do not alone give uniform constants as the underlying curve pinches.

The free-group construction of the moduli locus is the classical unitary Schottky construction [Flo01]. Daskalopoulos–Wentworth [DW93] analyze local degeneration of bundle moduli by weighted elliptic methods. Section 8 explains the hypotheses of the relevant kernel-continuation statements and proves the present relative Hodge-norm comparison directly by minimizing the energy of affine primitives. The uniform estimate does not require a lower bound for a nonzero eigenvalue of the vertex-gauge operator. Conformal changes of the surface metric alone do not distinguish the one-form energy.

For trivial bundles and a common neck scale, the actual noncommuting matrix-flow comparison was established in [Wue26]. We retain its full Chern-curvature equation and intrinsic trace comparison, and prove the weighted gluing estimates needed for independent node lengths and nontrivial unitary transport. The regular component correction at a node receives

contributions from every incident edge, whereas the node profile has its own logarithmic slope. Keeping both weights in the collar estimate gives the uniform residual $O(L_*^{-2}(1+\log L_*)^2)$. The trace comparison integrates this residual without a degeneration-dependent Sobolev constant or exponential time factor. For the all-time estimate we combine this finite-window comparison with exponential decay of both the exact and finite solutions. The lower bound for $L_*\nu_0$ makes these tails uniform in the rescaled time. The finite gradient structure does not identify the actual HYM equation with ordinary harmonic-map heat flow; its additional commutator term is retained in (23).

Finally, the spatial entry argument uses a fixed-component entropy: the logarithm of the determinant of the arithmetic matrix mean. It selects the constant limit metric with an exponential estimate, after which a global preparation metric transfers the fixed-component flows to the smoothing. Convergence at a fixed positive rescaled time also requires convergence of these component limits and the normalized finite parameters. Uniform tracking alone does not impose it.

For the family with varying component holonomy, a harmonic gauge on each punctured component separates its mean from a controlled remainder. The boundary error is bounded by the component energy, rather than by the size of its nearly parallel mean. Summing the disjoint component and neck energies then preserves arbitrary ratios of their scales. In the nonlinear comparison the curvature of a spatially constant component metric is canceled exactly by its finite component velocity. The additional matching error caused by component holonomy has order $\max_i |\alpha_i|/L$, which is small after integration on bounded intervals in s/L.

The inverse argument first uses component averages to identify the actual low spectral subspace with the vertex space. Polar normalization makes that identification isometric. The lower energy estimate gives a positive operator difference, and relative eigenvalue comparison bounds its trace; together these control the observed operator. On a known simple graph its off-diagonal blocks recover the adjoint transports. When adjacency is unknown, the block norms distinguish edges from nonedges. The recovered graph and the prescribed smoothing orders then supply the integral cycle pairing in the classical monodromy formula.

2. Compact smoothings and unitary gluing

We construct a compact family in which the six node parameters and the unitary gluing data can be varied independently. The metric is induced from a fixed smooth ambient Kähler metric. In particular its behaviour near a node differs from that of a hyperbolic collar.

2.1. **Six independent smoothing coordinates.** In $\mathbb{P}^2$ put

$$l_0 = x, \qquad l_1 = y, \qquad l_2 = z, \qquad l_3 = x + y + z, \qquad F_0 = \prod_{i=0}^{3} l_i.$$

The four lines have six distinct nodes p_{ij} and no triple intersection. For an unordered pair $e = \{i, j\}$ choose a nonzero constant γ_e and set

$$f_e = \gamma_e \left(\prod_{k \notin e} l_k \right)^2 .$$

Choose local frames of $\mathcal{O}_{\mathbb{P}^2}(4)$ near the six nodes and normalize $f_e(p_e) = 1$. At every other node p_f, the section f_e vanishes to order at least two. For $a = (a_e)$ near zero in $\mathbb{C}^6$, consider

$$\mathcal{C} = \left\{ ([x : y : z], a) : F_0 + \sum_e a_e f_e = 0 \right\} \subset \mathbb{P}^2 \times B. \tag{3}$$

Proposition 2.1. *After restricting B, the family (3) is proper, projective and flat, with smooth total space. There are holomorphic base coordinates $\tau = (\tau_e)$ centred at zero and disjoint node neighbourhoods in which the family is exactly*

$$u_e v_e = \tau_e. \tag{4}$$

Every fibre with all $\tau_e \neq 0$ is a smooth connected plane quartic of genus three.

Proof. At a central node the spatial Hessian of the local defining function is nondegenerate. Its derivative in the corresponding a_e direction is one. Elsewhere on the central fibre its spatial gradient is nonzero. The total hypersurface is therefore smooth near the entire central fibre; properness allows one restriction of the base for which it remains smooth.

Write the defining function near p_e as $f(z, a)$ in two affine coordinates. The implicit function theorem gives a unique nearby critical point $z_e(a)$. Its critical value $c_e(a) = f(z_e(a), a)$ satisfies

$$\frac{\partial c_e}{\partial a_f}(0) = f_f(p_e) = \delta_{ef}.$$

The derivative of z_e contributes nothing because the spatial gradient vanishes. The holomorphic Morse lemma with parameters then gives $f = c_e(a) + u_e v_e$. One may obtain these coordinates by translating the critical point, writing the remaining function using its integral Hessian, and taking a local holomorphic congruence square root of that nondegenerate symmetric matrix. This construction depends holomorphically on a and leaves the base fixed. The inverse function theorem now makes $\tau_e = -c_e(a)$ simultaneous base coordinates. Restrict to a polydisc in these coordinates.

The node charts have no singular point in a fibre with all $\tau_e \neq 0$. On the compact region outside them the spatial gradient remains nonzero. Properness and projectivity follow from (3). For flatness, its relative Cartier sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2 \times B}(-4) \longrightarrow \mathcal{O}_{\mathbb{P}^2 \times B} \longrightarrow \mathcal{O}_{\mathcal{C}} \longrightarrow 0$$

remains exact after restriction to every base fibre: none of the defining quartics is the zero polynomial. The local flatness criterion applies. Finally a smooth plane quartic is connected and has genus three. □

Give $\mathbb{P}^2$ its Fubini–Study form normalized so that a line has area one, and use the restriction to $\mathcal{C}$ of its sum with a positive Euclidean base form. The fibre metric is the induced Fubini–Study metric. Write dA_τ for its area form and C_i for the four central lines. Their limiting areas are $A_i = 1$. We retain A_i in formulas to exhibit the mass normalization; the arguments also apply to a different fixed smooth ambient Kähler metric.

Lemma 2.2. *For sufficiently small fixed node charts, the fibre metric in (4) is uniformly comparable to the metric induced from $\mathbb{C}^2$. The region $|u_e|, |v_e| < a$ has area at most Ca^2, independently of its annular modulus and of the other small parameters. On each fixed compact component exterior the complex structures and metrics converge smoothly.*

If $\rho > 0$, $b = \max_e |\tau_e| \ll \rho^2$, and the components are truncated at radius ρ, the coefficient errors in a fixed finite number of derivatives are bounded by $Cb\rho^{-N}$ for some fixed N.

Proof. On a fibre the other base coordinates are constant, and $v_e = \tau_e/u_e$. The induced Euclidean area is

$$\left(1 + \frac{|\tau_e|^2}{|u_e|^4}\right) dx\, dy, \qquad |\tau_e|/a < |u_e| < a.$$

Its integral is at most $2\pi a^2$. Smooth positivity of the ambient metric on the fixed total-space charts gives the uniform comparison. Away from the node centres the projection is a submersion; choose smooth component trivializations that preserve the branch coordinates on collars. Their coefficients converge smoothly. On a truncated collar, differentiating $v_e = \tau_e/u_e$ a fixed number of times costs only a fixed power of ρ^{-1}. Smooth dependence on the remaining parameters contributes $O(b)$ on fixed exteriors. Combining these estimates proves the last assertion. □

2.2. **The bundle and its actual holonomy.** Fix an integer $r \geq 2$. The edge set is that of K_4, and $U_{ij} \in \mathrm{SU}(r)$ denotes transport from the j frame to the i frame; set $U_{ji} = U_{ij}^{-1}$. This convention fixes the signs of all later incidence matrices.

Proposition 2.3. *Every tuple $U = (U_e) \in \mathrm{SU}(r)^6$ defines a holomorphic bundle E_U on a neighbourhood of the entire central fibre in $\mathcal{C}$, with a flat unitary connection D_U, parallel metric h_0, and a parallel trivialization of its determinant. Its component restrictions are trivial flat bundles. On a smooth fibre the holonomy factors through a surjection*

$$\pi_1(C_\tau) \longrightarrow \pi_1(K_4),$$

and graph-cycle holonomies are the ordered products of the specified edge maps.

Proof. Choose fixed $0 < a < b$ within the disjoint node charts and restrict $\max|\tau_e| < a^2$. Let the node open sets be $|u_e|, |v_e| < b$. Choose pairwise disjoint open component exteriors V_i whose incidence overlaps are the collars $a < |u_e| < b$ or $a < |v_e| < b$. Such sets follow from the smooth exterior trivializations used above. These trivializations need not be holomorphic: each open subset of the complex total space has its trivial holomorphic rank-r bundle. The two branch collars at a node are disjoint because $|u_e v_e| < a^2$. Thus there are no triple overlaps. Every incidence overlap is connected, being an annulus times the restricted base polydisc.

The sets cover the central fibre and, after a further restriction of the base, every entire fibre. Indeed a sequence of omitted points over parameters tending to zero would, by properness, have an omitted limit on the central fibre. Put constant special unitary transition matrices on the incidence overlaps, choosing their two values at an edge so that the induced component transport is U_{ij}. There is no nontrivial triple cocycle condition. The local connections d and identity metrics agree under these transitions. They give the asserted bundle, connection, metric and determinant trivialization.

On a smooth fibre the component pieces and node annuli are connected, as are their incidence overlaps. A graph walk is realized by paths within the component pieces and crossing paths in the annuli. Each closed graph walk therefore has a lift to a fibre loop. The van Kampen presentation obtained from this cover, after killing all patch loops, is the free fundamental group of the incidence graph, which is the barycentric subdivision of K_4. Patch loops have trivial transport in the specified flat frames. Their quotient consequently gives the stated surjection and the ordered transport formula. In particular each vanishing-cycle loop has trivial holonomy. □

By the unitary correspondence [NS65], $E_U|_{C_\tau}$ is polystable of degree zero, and it is stable exactly when the holonomy is irreducible. These are variations of holomorphic bundle structure and stabilizer, with fixed degree. No variation of the ordinary slope on a smooth curve is involved.

Example 2.4. Take $r = 2$, identity transports on $01, 02, 03, 23$, and

$$U_{12} = \frac{I + i\sigma_x}{\sqrt{2}}, \qquad U_{13} = \frac{I + i\sigma_y}{\sqrt{2}},$$

where $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices. The two cycle holonomies have no common complex invariant line, so every smooth-fibre bundle is stable. They cannot be removed by component frame changes. In the orthonormal basis $\sigma_a/\sqrt{2}$ of $\mathrm{Herm}_0(2)$ their adjoint maps are

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad R_y = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \tag{5}$$

Replacing the two holonomies by $\exp(i\alpha\sigma_x/2)$ and $\exp(i\beta\sigma_y/2)$ gives irreducible representations for small nonzero α, β, reducible ones on either axis, and trivial holonomy at the origin.

The gluing also has a holomorphic complex-parameter realization: use holomorphic $\mathrm{SL}(r, \mathbb{C})$ incidence matrices on complex parameter neighbourhoods. The analytic statements below concern the unitary real locus. They do not assert flatness in additional complex parameter directions.

2.3. **Scales and the finite quadratic form.** For a smooth fibre set

$$L_e = \log(1/|\tau_e|), \quad L_* = \min_e L_e, \quad w_e = \frac{2\pi}{L_e}, \quad w_* = \frac{2\pi}{L_*}. \tag{6}$$

The limit is $L_* \to \infty$, with no restriction on L_e/L_f or on the dependence of U on τ. The parameters remain fixed during every flow. On $\mathcal{V} = \bigoplus_{i=0}^{3} \mathrm{Herm}_0(r)$ put

$$M(X) = \sum_i A_i \operatorname{tr}(X_i^2), \qquad Q_{\tau,U}(X) = \sum_{e=ij} w_e \|X_i - U_{ij} X_j U_{ij}^*\|_{\mathrm{HS}}^2. \tag{7}$$

Each unoriented edge is counted once. Let $B_{\tau,U}$ be the self-adjoint operator for the inner product M whose quadratic form is $Q_{\tau,U}$, and let $\nu_0 \le \cdots \le \nu_{m-1}$ be its eigenvalues, where $m = 4(r^2 - 1)$. Zeros are included.

The kernel consists exactly of compatible tuples. Transport along a spanning tree determines such a tuple from its value at one vertex; the remaining edges require that value to commute with every cycle holonomy. Thus $\ker B_{\tau,U}$ is the trace-free Hermitian commutant of the representation. Its dimension may change with U, while the dimension m of the finite space is fixed.

3. The complete small spectrum

Let $\Delta_{\tau,U} = D_U^* D_U$ act on trace-free h_0-Hermitian endomorphisms over C_τ. We use the unnormalized area norm and the energy

$$\mathcal{Q}_{\tau,U}(s) = \int_{C_\tau} |D_U s|^2 \, dA_\tau.$$

Its eigenvalues $\lambda_0 \le \lambda_1 \le \cdots$ include the kernel. With the HYM convention in Section 5, this is the positive linearization at h_0.

Theorem 3.1. *For each fixed $r \geq 2$ there are $\gamma > 0$ and a function $\epsilon(\tau) \to 0$ as $L_* \to \infty$ such that, uniformly for $U \in \mathrm{SU}(r)^6$ and all positive length ratios,*

$$(1-\epsilon(\tau))\nu_k \leq \lambda_k \leq (1+\epsilon(\tau))\nu_k \quad (0 \leq k < m), \qquad \lambda_m \geq \gamma. \tag{8}$$

For sufficiently small parameters exactly m eigenvalues lie below a fixed positive threshold. The actual kernel is the space of compatible tuples in (7).

The multiplicative error in (8) remains meaningful when ν_k/w_* tends to zero at an arbitrary rate. Its proof uses the section energy to control every boundary error. An additive $o(w_*)$ bound would not imply this assertion.

3.1. **Trial sections.** Fix a small radius a independent of all parameters. The annulus $|\tau_e|/a < |u_e| < a$ has logarithmic length $\ell_e = L_e + 2\log a$. In its flat node frame, join X_i to $\mathrm{Ad}(U_{ij})X_j$ by affine interpolation in $\log|u_e|$ and keep the section constant on each component exterior. The resulting map $J_a : \mathcal{V} \to H^1$ is continuous across boundary circles; derivative jumps are allowed in the form domain. Conformal invariance of the two-dimensional Dirichlet energy gives the exact identity

$$\mathcal{Q}_{\tau,U}(J_a X) = \sum_e \frac{2\pi}{\ell_e}\|X_i - \mathrm{Ad}(U_{ij})X_j\|^2. \tag{9}$$

For fixed a and large L_*, $1 \leq L_e/\ell_e \leq 1 + C_a/L_*$. By Lemma 2.2, unitarity and smooth exterior convergence,

$$\big|\|J_a X\|_{L^2}^2 - M(X)\big| \leq \big(Ca^2 + o_a(\tau)\big)M(X), \tag{10}$$

where $o_a(\tau) \to 0$ for fixed a, uniformly in the unitary parameters. Min–max proves the relative upper bound with these errors, and gives $\lambda_{m-1} \leq Cw_*$.

3.2. **A weighted lower estimate.** For an arbitrary H^1 section s let $b_i(s)$ be its average, in the flat component frame, over the component exterior truncated at radius $a/2$. Let $c_{ie}(s)$ be its angular mean on the circle of radius a at an incidence. Fixed-domain Poincaré and trace inequalities give

$$\|c_{ie}(s) - b_i(s)\| \leq C_a \mathcal{Q}_{\tau,U}(s)^{1/2}. \tag{11}$$

The circle lies in the interior of the chosen truncated domain. Thus only uniformly smooth fixed collars are used in this estimate.

Angular averaging and Cauchy–Schwarz on each node annulus imply

$$\sum_e \widetilde{w}_e\|c_{ie} - \mathrm{Ad}(U_{ij})c_{je}\|^2 \leq \mathcal{Q}_{\tau,U}(s), \qquad \widetilde{w}_e = 2\pi/\ell_e.$$

Apply the triangle inequality in the direct sum of the edge spaces weighted by $\sqrt{\widetilde{w}_e}$. Equation (11), the fixed number of incidences and $\widetilde{w}_e \leq Cw_*$ give

$$\left(\sum_e \widetilde{w}_e\|b_i - \mathrm{Ad}(U_{ij})b_j\|^2\right)^{1/2} \leq (1 + C_a\sqrt{w_*})\mathcal{Q}_{\tau,U}(s)^{1/2}.$$

Since $w_e \leq \widetilde{w}_e$, we have proved

$$\mathcal{Q}_{\tau,U}(s) \geq (1 + C_a\sqrt{w_*})^{-2} Q_{\tau,U}(b(s)). \tag{12}$$

There is no reciprocal of the smallest edge weight in this estimate.

3.3. **Mass on the annuli.** We include the estimate which excludes additional small modes supported on the shrinking regions.

Lemma 3.2. *Let w be a vector-valued H^1 function on a node annulus with zero boundary values at $|u_e| = a$ and $|v_e| = a$. Then*

$$\|w\|_{L^2}^2 \le Ca^2 \int |dw|^2 \, dA_\tau,$$

with C independent of the annular modulus and all small parameters.

Proof. First use the induced Euclidean metric on the complex curve in $\mathbb{C}^2$ and put $f = |u_e|^2 + |v_e|^2$. Its coordinate functions are harmonic, so $\Delta_+ f = 4$ for the divergence-form operator $\Delta_+ = \operatorname{div} \nabla$, the negative of our nonnegative Laplacian. Also $|\nabla f| \le Ca$ on the annulus. Integration by parts, with zero boundary trace, yields

$$4 \int |w|^2 = -2 \int \langle w, dw \rangle \cdot \nabla f \le Ca \|w\|_{L^2} \|dw\|_{L^2}.$$

Squaring proves the estimate. The same argument applies to each real coordinate of w. Uniform comparison of areas in Lemma 2.2 and conformal invariance of the Dirichlet energy transfer it to the actual metric. □

Lemma 3.3. *If s belongs to the span of at most $m + 1$ orthonormal eigensections with eigenvalues at most $\Lambda \le 1$, then*

$$\big|M(b(s)) - \|s\|_{L^2}^2\big| \le \big(Ca^2 + C_a\sqrt{\Lambda} + o_a(\tau)\big)\|s\|_{L^2}^2. \tag{13}$$

All constants are uniform in U and in the node-length ratios.

Proof. Start with a normalized eigensection of eigenvalue $\lambda \le 1$. On a fixed component exterior, Poincaré gives $\|s - b_i\|_{L^2} \le C_a\sqrt{\lambda}$. In the flat frame the equation is the scalar eigenfunction equation in each real coordinate. Interior $W^{2,2}$ estimates on slightly smaller fixed collars, followed by the two-dimensional embedding into C^0, give the same order on each boundary circle. The averages themselves are uniformly bounded by Cauchy–Schwarz.

On a node annulus write $s = h + w$, where h is the coordinatewise harmonic extension of its boundary values and w has zero boundary trace. The maximum principle bounds h by the boundary supremum, and the annular area is $O(a^2)$. Orthogonality in Dirichlet energy gives $E(w) \le E(s) \le \lambda$; Lemma 3.2 bounds its mass. Combining the component and annular estimates yields (13) for one eigenfunction. The difference between the radii $a/2$ used for averaging and a used to partition the surface is covered by the same component Poincaré estimate.

For a unit linear combination of at most $m+1$ orthonormal eigensections, sum the boundary and exterior estimates and use Cauchy–Schwarz in the finite coefficient space. Its energy is at most Λ. The preceding harmonic-extension argument then applies without change, with a constant depending on m. □

Proof of Theorem 3.1. If λ_m tended to zero along any sequence with $L_* \to \infty$, choose a small and then use Lemma 3.3 on its first $m + 1$ eigensections. Averaging would be injective from an $(m + 1)$-dimensional space to $\mathcal{V}$, a contradiction. Thus $\lambda_m \ge \gamma > 0$ uniformly. The upper bound $\lambda_{m-1} \le Cw_* \to 0$ identifies precisely the asserted cluster.

On the span of the first $k + 1$ eigensections, $k < m$, take $\Lambda = Cw_*$. Averaging is injective and almost isometric. Finite min–max supplies a nonzero s in that span for which

$Q_{\tau,U}(b(s)) \geq \nu_k M(b(s))$. Therefore

$$\lambda_k \geq \frac{\mathcal{Q}_{\tau,U}(s)}{\|s\|_{L^2}^2} \geq (1 + C_a\sqrt{w_*})^{-2}\big(1 - Ca^2 - C_a\sqrt{w_*} - o_a(\tau)\big)\nu_k.$$

Together with (9)–(10) this proves the relative comparison: first choose a for a prescribed error and then take all parameters sufficiently small. This order gives a single uniform error modulus tending to zero. No division by ν_k or by a smallest weight has occurred.

Finally zero energy means that a section is parallel. It is constant in every flat patch, and its values obey the edge compatibility relations. Conversely each compatible tuple extends to such a section. This proves the exact kernel assertion even when its dimension changes. □

Corollary 3.4. *Let $\Pi_{\tau,U}$ project orthogonally onto the first m eigensections. On its orthogonal complement,*

$$\|\Delta_{\tau,U}^{-1}\|_{L^2\to L^2} \leq \gamma^{-1}, \qquad \|e^{-s\Delta_{\tau,U}}(1 - \Pi_{\tau,U})\|_{L^2\to L^2} \leq e^{-\gamma s}.$$

For $z \notin [\gamma, \infty)$ the complementary resolvent has norm at most $\operatorname{dist}(z, [\gamma, \infty))^{-1}$.

Proof. These are the spectral-calculus bounds for the nonnegative self-adjoint operator on that subspace. □

These are L^2 estimates. They do not assert a uniform global Schauder or Sobolev estimate on the degenerating surface.

3.4. **A compact example with three weak scales.** Use the rank-two transports in Example 2.4 and the Fubini–Study normalization $A_i = 1$. For $h > 0$ put

$$w_{01} = w_{02} = w_{03} = h, \qquad w_{12} = h^2, \quad w_{23} = h^3, \quad w_{13} = h^4, \qquad \tau_e(h) = e^{-2\pi/w_e(h)}.$$

This is a real path in the complex family of Proposition 2.1; it is not claimed to extend as a holomorphic disc at $h = 0$. Every sufficiently small positive h gives a stable bundle on a smooth compact genus-three curve.

Proposition 3.5. *The twelve weak eigenvalues along this path satisfy*

$$\begin{aligned} \lambda_0 \sim \tfrac{1}{2}h^4, \qquad \lambda_1, \lambda_2 \sim \tfrac{1}{2}h^2, \\ \lambda_3, \dots, \lambda_8 \sim h, \qquad \lambda_9, \lambda_{10}, \lambda_{11} \sim 4h, \qquad \lambda_{12} \geq \gamma > 0. \end{aligned} \tag{14}$$

Each member of a displayed group has the stated asymptotic.

Proof. In three-by-three blocks the finite mass matrix is I_{12} and the energy matrix is

$$Q_h = \begin{pmatrix} 3hI & -hI & -hI & -hI \\ -hI & (h + h^2 + h^4)I & -h^2R_x & -h^4R_y \\ -hI & -h^2R_x^T & (h + h^2 + h^3)I & -h^3I \\ -hI & -h^4R_y^T & -h^3I & (h + h^3 + h^4)I \end{pmatrix}. \tag{15}$$

The order-h star form has a three-dimensional kernel $K = \{(x, x, x, x)\}$ and positive eigenvalues 1 with multiplicity six and 4 with multiplicity three. Thus nine eigenvalues have the asserted order-h coefficients. The orthonormal embedding of K is $x \mapsto (x, x, x, x)/2$, and compression of the order-h^2 coefficient is

$$\tfrac{1}{4}(I - R_x)^T(I - R_x) = \operatorname{diag}(0, \tfrac{1}{2}, \tfrac{1}{2}).$$

Trials in K bound the remaining three eigenvalues by $O(h^2)$. At such spectral parameters the $K^\perp$ block is $hA + O(h^2)$ with $A > 0$, and its inverse is $O(h^{-1})$. The cross block is $O(h^2)$,

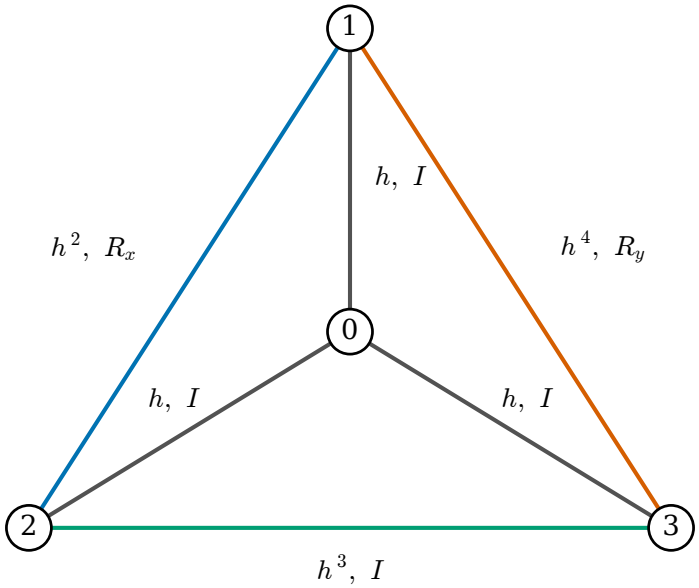


FIGURE 1. The actual attachment graph for the three-scale family. Each edge is labelled by its capacity weight and adjoint transport. The displayed transport on an edge ij, $i<j$, is from j to i. The matrices R_x, R_y are defined in (5).

so its Schur contribution is $O(h^3)$. Two eigenvalues are therefore asymptotic to $h^2/2$, and one is $o(h^2)$.

The matrix is positive definite for $h>0$: zero energy first forces a common vertex value, which must be fixed by both R_x and R_y; their fixed axes meet only at zero. Direct expansion of the printed matrix gives

$$\begin{aligned}\det Q_h = 4h^{17}\big(&2+4h+14h^2+36h^3+55h^4+118h^5+145h^6\\&+194h^7+225h^8+178h^9+177h^{10}+78h^{11}+54h^{12}\big).\end{aligned}\tag{16}$$

The product of the nine fast eigenvalues is asymptotic to $64h^9$, and that of the two middle eigenvalues to $h^4/4$. Dividing the determinant by these products gives $h^4/2$ for the remaining eigenvalue. The relative comparison (8) transfers all these coefficients to the actual operator. □

The accompanying exact calculation also verifies the three rescaled characteristic-polynomial limits. For $P(\lambda,h)=\det(\lambda I-Q_h)$ they are

$$\begin{aligned}h^{-12}P(hz,h)&\longrightarrow z^3(z-1)^6(z-4)^3,\\h^{-15}P(h^2z,h)&\longrightarrow -16z(2z-1)^2,\\h^{-17}P(h^4z,h)&\longrightarrow -8(2z-1).\end{aligned}$$

These checks agree with the compression and determinant argument above.

The inverse eigenvalue scales in (14) are linearized relaxation times. In particular, the weakest scale is not a nonlinear tracking conclusion. In this example $L_* = 2\pi/h$ and $L_*\nu_0 \sim \pi h^3 \to 0$, so the all-time hypothesis in Theorem 5.3 below fails.

4. The finite metric evolution

Let $\mathcal{P}_r$ be the positive Hermitian determinant-one matrices, with metric $g_P(V,W) = \operatorname{tr}(P^{-1}VP^{-1}W)$. Its distance and geodesic are

$$\begin{aligned} d(P,Q)^2 &= \operatorname{tr}\Big(\log(P^{-1/2}QP^{-1/2})^2\Big), \\ \operatorname{Geo}(P,Q;\theta) &= P^{1/2}\exp\Big(\theta\log(P^{-1/2}QP^{-1/2})\Big)\,P^{1/2}. \end{aligned} \tag{17}$$

This is a complete simply connected symmetric space of nonpositive curvature; see [Bha13, Sections 2.3–2.6]. Unitary congruences act isometrically and fix I. On $\mathcal{P}_r^4$ use the product metric with weights A_i. With physical weights (6), define

$$\Phi_{\tau,U}(\mathbf{H}) = \frac{1}{2}\sum_{e=ij} w_e d(H_i, U_{ij}H_jU_{ij}^*)^2. \tag{18}$$

Proposition 4.1. *The negative gradient flow of (18) is*

$$H_i^{-1}\dot{H}_i = \frac{1}{A_i}\sum_{e=ij} w_e B_{ij}, \qquad B_{ij} = \log(H_i^{-1}U_{ij}H_jU_{ij}^*). \tag{19}$$

For initial tuples in a fixed bounded ball, it exists for all physical times $s \geq 0$ and remains in a compact matrix range independent of τ, U. Distances between two such solutions with the same parameters are nonincreasing. On that range the vector field and every fixed-order tuple derivative are $O(L_*^{-1})$, uniformly in all length ratios and U.

Proof. The gradient of $\frac{1}{2}d(P,Q)^2$ with respect to P is $-\operatorname{Log}_P Q = -P\log(P^{-1}Q)$. Summing the incident terms and dividing by A_i gives (19). Each B_{ij} is H_i-self-adjoint and trace free, so the velocity is tangent to $\mathcal{P}_r$. The reverse identity is $B_{ji} = -U_{ij}^*B_{ij}U_{ij}$; no commutation of the endpoint metrics is used.

Squared distance is jointly geodesically convex on this Hadamard manifold. Hence Φ is convex and vanishes at $\mathbf{I} = (I,I,I,I)$. If D_A is the weighted product distance, gradient-flow convexity gives

$$\frac{d}{ds}\frac{1}{2}D_A(\mathbf{H},\mathbf{I})^2 \leq -\Phi(\mathbf{H}). \tag{20}$$

Closed bounded subsets are compact, so the smooth vector field continues for all time. Applying the same convexity inequality at the two endpoints of the joining geodesic proves nonexpansion between two solutions. Finally (19) is a finite sum of smooth matrix functions with coefficients $w_e \leq 2\pi/L_*$. All tuple derivatives are bounded on the confined range, uniformly in the compact unitary parameter space. In particular $|\dot{B}_{ij}| \leq C/L_*$. □

The zero set of Φ consists of compatible metric tuples. On irreducible unitary representations this set is $\{\mathbf{I}\}$ by Schur's lemma and determinant one. On reducible representations it is generally positive-dimensional.

Lemma 4.2. *Put $S_i = \log H_i$ and $R_f = \sum_i A_i\|S_i\|^2$. Then*

$$\Phi(\mathbf{H}) \geq \tfrac{1}{2}Q_{\tau,U}(S) \geq \tfrac{1}{2}\nu_0 R_f, \qquad R_f(s) \leq e^{-\nu_0 s}R_f(0).$$

Proof. The exponential map at I expands distances in this nonpositively curved symmetric space. Since unitary congruence commutes with the exponential, $d(H_i, UH_jU^*) \geq \|S_i - US_jU^*\|$. This proves the energy inequalities. Moreover $R_f = D_A(\mathbf{H}, \mathbf{I})^2$, so (20) gives $R_f' \leq -2\Phi \leq -\nu_0 R_f$. □

The finite equation has the same linearization $\dot{X} = -B_{\tau,U}X$ as the effective spectral operator. Its nonlinear variables need not commute. For Example 2.4, no fixed simultaneous diagonalization of the cycle transports reduces the equation to independent scalar vertex equations.

5. Reconstruction and the actual HYM flow

The base and bundle parameters are fixed throughout this section. All exact initial metrics are smooth, have the parallel determinant, and obey fixed positive bounds against h_0. In local unitary flat frames, these conditions mean $\det H = 1$ and $cI \leq H \leq CI$. We use physical time s and the normalization

$$H^{-1}\partial_s H = -2i\Lambda_{\omega_\tau} F_H. \tag{21}$$

For $\omega_\tau = (i/2)a\, dz \wedge d\bar{z}$ this reads

$$\partial_s H = 4a^{-1}(H_{z\bar{z}} - H_{\bar{z}}H^{-1}H_z). \tag{22}$$

The scalar nonnegative Laplacian is $\Delta_\tau = -4a^{-1}\partial_z\partial_{\bar{z}}$. The real-coordinate expression is

$$\begin{aligned} a\partial_s H = {} & H_{xx} + H_{yy} - H_x H^{-1} H_x - H_y H^{-1} H_y \\ & + i(H_x H^{-1} H_y - H_y H^{-1} H_x). \end{aligned} \tag{23}$$

The last term need not vanish. We retain it in every curvature calculation.

For two metrics put

$$\sigma(H, K) = \operatorname{tr}(H^{-1}K + K^{-1}H) - 2r.$$

This is intrinsic under the bundle transition maps. If e^{b_j} are the eigenvalues of $H^{-1}K$, then $d(H,K)^2 = \sum_j b_j^2 \leq \sum_j (e^{b_j} + e^{-b_j} - 2) = \sigma(H,K)$. For a smooth comparison metric define

$$\mathcal{R}(K) = K^{-1}\partial_s K + 2i\Lambda F_K, \tag{24}$$

measured in the operator norm induced by K.

Lemma 5.1. *The exact solution of (21) exists for all $s \geq 0$ and preserves its determinant and bounds $ch_0 \leq H \leq Ch_0$. For any smooth comparison $K(s)$ with common positive bounds, one has*

$$\int \sigma(H(s), K(s)) \leq \int \sigma(H(0), K(0)) + C_0 \int_0^s \int |\mathcal{R}(K(v))|_{\mathrm{op}}\, dA_\tau\, dv. \tag{25}$$

Here and below unlabelled spatial integrals use dA_τ. The constant depends only on rank and the positive bounds, not on a spectral gap or Sobolev constant.

Proof. Smooth all-time existence on a fixed compact base is the classical metric heat-flow result [Don85]; our normalization and order preservation are also given in [HKKP18, Section 4.1 and Proposition 4.3]. Apply order preservation to the stationary metrics ch_0, Ch_0. The logarithm of the determinant satisfies the scalar heat equation, so its constant initial value is preserved.

For completeness the local Donaldson trace calculation has the forced form, with $q = H^{-1}K$,

$$(\partial_s + \Delta_\tau)\sigma(H, K) \leq \operatorname{tr}((q - q^{-1})\mathcal{R}(K)).$$

For two exact velocities choose a holomorphic frame normal for H at the point. The curvature and Laplacian terms in the forward trace cancel, leaving $-4a^{-1}\operatorname{tr}(K_{\bar z}K^{-1}K_z)\le 0$. Interchange the metrics for the inverse trace. The additional velocity $K\mathcal{R}(K)$ contributes $\operatorname{tr}(q\mathcal{R})$ and $-\operatorname{tr}(\mathcal{R}q^{-1})$, respectively. Both q and $\mathcal{R}$ are K-self-adjoint, and the common bounds control $\operatorname{tr} q+\operatorname{tr} q^{-1}$. This proves the scalar inequality with right-hand side at most $C_0|\mathcal{R}|_{\mathrm{op}}$. Integration on the whole closed fibre removes the Laplacian and gives (25). The calculation is unchanged by constant unitary transition maps. □

Write $\varepsilon_e=L_e^{-1}$, $\varepsilon_*=L_*^{-1}$, and

$$e_*=\frac{(1+\log L_*)^2}{L_*}. \tag{26}$$

The initial tuples for (19) lie in a fixed bounded ball.

Theorem 5.2. *For all sufficiently large L_* there is a smooth determinant-one comparison metric $\widehat K(s)$, defined for all $s\ge 0$ from the finite trajectory (19), with uniform positive bounds and*

$$\sup_{s\ge 0}\int|\mathcal{R}(\widehat K(s))|_{\mathrm{op}}\le CL_*^{-2}(1+\log L_*)^2. \tag{27}$$

On every fixed compact subset of component i, $d(\widehat K(s),H_i(s))\le C_K/L_*$ uniformly in s. If an exact flow has initial discrepancy $\eta=\int\sigma(H^{\mathrm{in}},\widehat K(0))$, then, for every $T\ge 0$,

$$\sup_{0\le s\le TL_*}\int d(H(s),\widehat K(s))^2\le \eta+CTe_*. \tag{28}$$

The constants are independent of T, all positive length ratios and all unitary parameters. Uniform convergence over a class of initial metrics requires uniform $\eta\to 0$.

5.1. **Component potentials.** Let g_{ie} be the mean-zero scalar Green function on the fixed closed component C_i, normalized by

$$\Delta_i g_{ie}=-2\pi\delta_{p_{ie}}+\frac{2\pi}{A_i},\qquad g_{ie}=\log|u_e|+\gamma_{ie}(u_e)$$

near the incidence. The regular part γ_{ie} is smooth. Define

$$h_i=-\sum_{e=ij}\varepsilon_e B_{ij}g_{ie},\qquad K_i=H_i\exp(h_i). \tag{29}$$

The logarithm h_i is trace free and H_i-self-adjoint. Away from the punctures,

$$\Delta_i h_i=-H_i^{-1}\dot H_i. \tag{30}$$

Cut the components at radius $\rho=\varepsilon_*$. Since $\max|\tau_e|=e^{-L_*}\ll\rho^2$, the two branches of each node are disjoint there. Fixed Green-function estimates and Proposition 4.1 give

$$\begin{aligned}&\|h_i\|_\infty\le C\varepsilon_*(1+|\log\rho|),\quad \int|h_i|\le C\varepsilon_*,\qquad \int|h_i|^2\le C\varepsilon_*^2,\\ &\int|\nabla h_i|^2\le C\varepsilon_*^2(1+|\log\rho|),\quad \int|\dot h_i|\le C\varepsilon_*^2,\quad \int|h_i||\Delta_i h_i|\le C\varepsilon_*^2.\end{aligned} \tag{31}$$

In particular the integral bounds use integrability of $\log|u|$ and its square, not their suprema on the truncated domain. The time derivative estimate uses $|\dot B_{ij}|\le C\varepsilon_*$ and $\varepsilon_e\le\varepsilon_*$.

The exact exponential identity

$$e^{-h}\partial e^{h} = \int_0^1 e^{-th}(\partial h)e^{th}\,dt$$

and its time analogue show, after (30), that

$$|\mathcal{R}(K_i)| \le C\big(|\dot h_i| + \varepsilon_*|h_i| + |\nabla h_i|^2 + |h_i||\Delta_i h_i|\big).$$

Indeed the linear spatial term is $\Delta_i h_i$; all terms produced by differentiating the conjugations contain either two first derivatives or h_i times a second derivative. The time conjugation of $H_i^{-1}\dot H_i$ contributes $O(\varepsilon_*|h_i|)$. These identities include the commutators in (23). Integrating (31) bounds the residual by $C\varepsilon_*^2(1+|\log\rho|)$ on the fixed component. Transfer to the actual exterior costs $Cb\rho^{-N}$, where $b = e^{-L_*}$, by Lemma 2.2. A fixed N suffices for the finite derivative order involved.

5.2. **Node profiles and mixed weights.** Near an incidence write

$$h_i = -\varepsilon_e B_{ij}\log|u_e| + c_{ie} + q_{ie}(u_e), \qquad q_{ie}(0) = 0.$$

The regular terms include contributions from every incident edge. Uniformly on the collar, $|c_{ie}| \le C\varepsilon_*$ and $|D^k q_{ie}| \le C\varepsilon_*\rho^{1-k}$ for $k \le 2$. Their time derivatives gain a factor ε_*. Set $P_{ie} = H_i\exp(c_{ie})$. In the i frame the node metric is

$$N_e(s,u_e) = \mathrm{Geo}(P_{ie}, U_{ij}P_{je}U_{ij}^*; -\varepsilon_e\log|u_e|). \tag{32}$$

The reverse branch has parameter $1+\varepsilon_e\log|u_e|$, because $u_e v_e = \tau_e$. Geodesic reversal and equivariance under unitary congruence therefore give exactly the same metric in the other frame. The individual ε_e is essential for this compatibility.

At fixed time a geodesic in (32) is $P\exp(\theta B)$ with a fixed matrix B and harmonic scalar θ. Its Chern curvature is zero on the annulus. Its time derivative is $O(\varepsilon_*)$ on the bounded endpoint range; the middle region has area $O(\rho^2)$, so its integrated time residual is $O(\varepsilon_*\rho^2)$.

On $\rho \le |u_e| \le 2\rho$ use relative logarithms $\ell_i(K) = \log(H_i^{-1}K)$. For fixed endpoint metrics H_i, H_j, and small self-adjoint endpoint logarithms C_i, C_j, define the difference

$$\mathcal{D}(C_i,C_j;\theta) = \ell_i\big(\mathrm{Geo}(H_ie^{C_i}, U_{ij}H_je^{C_j}U_{ij}^*;\theta)\big) - C_i - \theta B_{ij}.$$

Self-adjointness here is with respect to H_i and H_j, respectively. With no endpoint correction the node logarithm is exactly θB_{ij}, so $\mathcal{D}(0,0;\theta) = 0$. At $\theta = 0$ the geodesic equals $H_ie^{C_i}$, so $\mathcal{D}(C_i,C_j;0) = 0$. The geodesic and logarithm are smooth on the confined endpoint range. Integrating their bounded mixed derivatives first along $q(C_i,C_j)$, $0 \le q \le 1$, and then from 0 to θ gives

$$|\mathcal{D}(C_i,C_j;\theta)| \le C(|C_i|+|C_j|)|\theta|.$$

Its first two θ derivatives are bounded by $C(|C_i|+|C_j|)$, since they also vanish when both corrections are zero.

Apply this with $C_i = c_{ie}$, $C_j = c_{je}$ and $\theta = -\varepsilon_e\log|u_e|$. The endpoint corrections are $O(\varepsilon_*)$, whereas the spatial derivatives of θ have size $O(\varepsilon_e\rho^{-k})$ for $k = 1,2$. Thus the correction to the node logarithm is bounded, through two spatial derivatives, by $C\varepsilon_*\varepsilon_e(1+|\log\rho|)\rho^{-k}$. The component logarithm differs from $c_{ie}+\theta B_{ij}$ by the regular term q_{ie}, whose corresponding bound is $C\varepsilon_*\rho^{1-k}$. Combining the two estimates gives

$$|D^k(\ell_i(K_i) - \ell_i(N_e))| \le C\left(\varepsilon_*\rho^{1-k} + \varepsilon_*\varepsilon_e(1+|\log\rho|)\rho^{-k}\right), \qquad k = 0,1,2. \tag{33}$$

The endpoint correction uses all incident weights, whereas the node slope uses ε_e. No inverse of ε_e enters.

Join these logarithms with a fixed rescaled radial cutoff and exponentiate relative to H_i. This preserves positivity and determinant one. If n is the node logarithm and δ is its cutoff difference from the component logarithm, the full curvature difference is bounded by

$$C\big(|D^2\delta| + |Dn||D\delta| + |\delta|(|D^2n| + |Dn|^2) + |D\delta|^2\big).$$

Here $|Dn| \le C\varepsilon_e/\rho$ and $|D^2n| \le C\varepsilon_e/\rho^2$. Multiply by the collar area $O(\rho^2)$ and substitute (33). The integrated curvature error is at most

$$C\big(\varepsilon_*\rho + \varepsilon_*\varepsilon_e(1 + |\log\rho|)^2\big).$$

Time terms contribute $O(\varepsilon_*\rho^2)$. Terms of higher order are absorbed since $\varepsilon_*|\log\rho| \to 0$. The same geometric transfer estimate contributes $Cb\rho^{-N}$.

Proof of Theorem 5.2. Use (29), (32) and the logarithmic cutoffs to define one global $\widehat{K}(s)$ for the entire finite trajectory. Their agreement under the actual transition matrices proves that this is a metric on E_U, not just a collection of matrices. The confined endpoint range and the small component logarithms give uniform positive bounds. On each fixed component compact subset $h_i = O(\varepsilon_*)$.

Sum the component, middle-node and collar estimates with $\rho = \varepsilon_*$. The exponentially small $e^{-L_*}L_*^N$ is absorbed by $L_*^{-2}(1+\log L_*)^2$. All constants are uniform for $s \ge 0$, because the finite trajectory stays in one compact range. This proves (27). Lemma 5.1 on the whole closed fibre then gives (28) after integration to TL_* and the inequality $d^2 \le \sigma$. □

5.3. **A uniform all-time conclusion.**

Theorem 5.3. *Under the hypotheses of Theorem 5.2, suppose in addition that*

$$L_*\nu_0 \ge \kappa > 0. \tag{34}$$

Then, for sufficiently large L_*,

$$\sup_{s\ge0}\int d(H(s),\widehat{K}(s))^2 \le C\eta + C_\kappa e_*(1+|\log e_*|).$$

This is absolute tracking in the indicated norm.

Proof. For the exact solution set $\mathcal{F}(H) = \int(\operatorname{tr}(h_0^{-1}H) - r)$. Since $S = \log(h_0^{-1}H)$ is trace free and the metrics have fixed positive bounds, $\mathcal{F}$ is uniformly comparable to $\int|S|^2 = \int d(H,h_0)^2$. Taking the trace of (22) and integrating gives

$$-\mathcal{F}' = 4\int \operatorname{tr}(H_{\bar z}H^{-1}H_z)\,dx\,dy \ge c_1\int |D_UH|^2\,dA_\tau.$$

Here $H_{\bar z} = H_z^*$. Replace H^{-1} by its uniform lower bound before using $4\operatorname{tr}(H_z^*H_z) = \operatorname{tr}(H_x^2 + H_y^2)$. This does not discard a weighted commutator. The derivative of the matrix logarithm is bounded on the common matrix range, and hence $|D_US| \le C|D_UH|$. Theorem 3.1 and (34) imply $\lambda_0 \ge \kappa/(2L_*)$ and therefore

$$-\mathcal{F}' \ge c_2\lambda_0\mathcal{F}, \qquad \int d(H(L_*\sigma),h_0)^2 \le Ce^{-c_3\kappa\sigma}.$$

Lemma 4.2 supplies the corresponding exponential tail for the finite trajectory. The reconstruction vanishes at $\mathbf{H} = \mathbf{I}$ and is uniformly Lipschitz from the confined tuple range to global L^2: on components this follows from the Green construction and on nodes from smoothness of bounded geodesics. Its squared distance from h_0 has the same type of exponential tail. For $0 \le s \le TL_*$ use (28); for $s \ge TL_*$ use the two tails and the triangle inequality. Choosing $T = C_\kappa(1+|\log e_*|)$ proves the estimate. □

The condition (34) permits some arbitrarily weak edges. In rank two keep $01, 02, 03, 12, 13$ comparable to w_*, with the two noncommuting transports of Example 2.4, and let 23 have any smaller positive weight. The first five edges already have zero trace-free kernel and a positive normalized gap; the last nonnegative term cannot decrease it. In contrast, Proposition 3.5 does not satisfy (34). On its weakest time h^{-4} the accumulated bound in (28) is not small. Uniform nonlinear tracking on that time requires an additional argument.

6. Spatial initial metrics and the initial layer

The initial-error hypothesis of Theorem 5.2 can be obtained from spatially varying metrics on the normalization components. We first select the constant metric determined by the actual flow on each fixed component. We then construct an initial layer on the smooth fibre whose duration is negligible compared with L_*.

6.1. **Constant limits on a fixed component.** Let Y be a fixed connected compact Riemann surface of area A, with its trivial holomorphic rank-r bundle and fixed flat frame. Write $\langle f\rangle = A^{-1}\int_Y f\,dA$, entrywise for a matrix. Let $\lambda_1 > 0$ be the first positive scalar eigenvalue. When a component flow converges to a constant metric P, we call P the asymptotic phase of its initial metric.

Theorem 6.1. *For a smooth determinant-one initial metric with $cI \le H^{\mathrm{in}} \le CI$, $0 < c \le 1 \le C$, the actual HYM flow converges to a unique constant matrix $P = P(H^{\mathrm{in}}) \in \mathcal{P}_r$. Set*

$$M(s) = \langle H(s)\rangle, \quad J(s) = \log\det M(s), \quad V(s) = \langle \|H(s) - M(s)\|^2_{\mathrm{HS}}\rangle, \quad \gamma = 2c^2C^{-2}\lambda_1.$$

Then M decreases in Loewner order to P, and

$$\frac{V}{2C^2} \le J \le \frac{V}{2c^2}, \quad J(s) \le J(0)e^{-\gamma s}, \quad 0 \preceq M(s) - P, \quad \operatorname{tr}(M(s) - P) \le CJ(s). \tag{35}$$

In particular

$$\langle \|H(s) - P\|^2_{\mathrm{HS}}\rangle \le C^2(2J(0) + J(0)^2)e^{-\gamma s}. \tag{36}$$

For $s \ge 1$ the same exponential order bounds $\sup_Y d(H(s), P)^2$, with a constant depending only on Y, c, C, r. For two such initial metrics,

$$A\,\sigma(P(H^{\mathrm{in}}), P(K^{\mathrm{in}})) \le \int_Y \sigma(H^{\mathrm{in}}, K^{\mathrm{in}})\,dA. \tag{37}$$

Proof. Equation (22) is $H_s = -\Delta H - \mathcal{A}(H)$, with $\mathcal{A}(H) = 4a^{-1}H_z^*H^{-1}H_z \succeq 0$. Its spatial mean therefore satisfies $M' = -\langle\mathcal{A}(H)\rangle \preceq 0$. Since $cI \le M \le CI$, it has a positive matrix limit P. At this point its determinant has not been prescribed.

For a Hermitian direction Z and a matrix $cI \le B \le CI$,

$$-D^2(\log\det)_B[Z, Z] = \operatorname{tr}(B^{-1}ZB^{-1}Z), \qquad C^{-2}\|Z\|^2 \le \operatorname{tr}(B^{-1}ZB^{-1}Z) \le c^{-2}\|Z\|^2.$$

Diagonalizing B proves the inequalities entrywise. Apply Taylor's integral remainder along the segment from M to $H(x)$, then average. The linear term vanishes and $\log\det H(x) = 0$, giving the two bounds for J in (35).

Using positivity twice gives

$$-J' = \operatorname{tr}(M^{-1}\langle\mathcal{A}(H)\rangle) \ge C^{-1}\langle\operatorname{tr}\mathcal{A}(H)\rangle \ge C^{-2}\langle|dH|^2\rangle.$$

In the last step one first bounds H^{-1} below and then uses the unweighted identity $4a^{-1}\operatorname{tr}(H_z^*H_z) = |dH|^2$. The scalar Poincaré inequality in each matrix coordinate gives $\langle|dH|^2\rangle \ge \lambda_1 V$. Thus $J' \le -\gamma J$. In particular $J \to 0$, so continuity of determinant gives $\det P = 1$.

The segment from P to M is bounded above by CI. Since $M - P \succeq 0$,

$$J = \int_0^1 \operatorname{tr}\big[(P + t(M-P))^{-1}(M-P)\big]\, dt \geq C^{-1} \operatorname{tr}(M-P).$$

The mean-zero cross term in the squared norm vanishes, and therefore

$$\langle \|H - P\|^2 \rangle = V + \|M - P\|^2 \leq 2C^2 J + C^2 J^2.$$

This proves (36).

The constant P is itself an exact solution. The scalar trace comparison shows that $\sigma(H(s), P)$ is a nonnegative heat subsolution. On the fixed component the time-one heat kernel is bounded, whence

$$\sup_Y \sigma(H(s), P) \leq C_Y \int_Y \sigma(H(s-1), P)\, dA \quad (s \geq 1).$$

On the fixed matrix range σ is comparable to squared matrix difference, so (36) proves uniform convergence with the asserted rate. Finally apply trace contraction to two exact flows and pass to their uniform constant limits. This gives (37). □

The phase depends on the entire spatial initial metric. For diagonal initial data $\operatorname{diag}(e^{f_1}, \dots, e^{f_r})$, $\sum f_j = 0$, it is $\operatorname{diag}(e^{\langle f_1 \rangle}, \dots, e^{\langle f_r \rangle})$, by scalar heat flow. No formula $P = \exp\langle \log H^{\mathrm{in}} \rangle$ is asserted for general noncommuting data. The contraction estimate makes the phase map Lipschitz in L^2 on every fixed bounded metric range.

6.2. **An open class on the smooth fibres.** Put $\rho = L_*^{-1}$ and let $\Omega_i(2\rho) \subset C_i$ be the complement of the incidence discs of radius 2ρ. Use the fixed exterior transfers and flat frames of Section 2. For a smooth trace-free h_0-Hermitian endomorphism S on C_τ, fix $0 < \beta < 1$ and define

$$\mathcal{N}_\tau(S) = \|S\|_{C^0(C_\tau)} + \inf_{\widetilde{S}_i|_{\Omega_i(2\rho)} = S_i} \sum_i \|\widetilde{S}_i\|_{C^{6,\beta}(C_i)}. \tag{38}$$

The extensions are trace-free Hermitian matrices in the component frames. Fix $R_0 > 0$ and take $\mathcal{N}_\tau(S) < R_0$, with initial metric $H^{\mathrm{in}} = h_0 e^S$. For each fixed smooth fibre this class is open in the $C^{6,\beta}$ topology on smooth metrics: fixed-domain extension operators bound $\mathcal{N}_\tau(S_1 - S_2)$ by a finite multiple of the global $C^{6,\beta}$ norm. Their operator norms need not be uniform as $\rho \to 0$. The radius in (38) is fixed independently of the degeneration. Small fixed component profiles joined smoothly on the nodes give nonempty examples. Bounded smooth changes inside the middle necks may have arbitrary derivatives.

Choose extensions with total norm less than $2R_0$. Apply Theorem 6.1 to $e^{\widetilde{S}_i}$ and denote its flow by $G_i(s, z)$ and selected phase by P_i. All component flows have uniform positive bounds and uniform exponential phase convergence. Different admissible extensions agree outside discs of area $O(\rho^2)$, so (37) gives $d(P_i, \widetilde{P}_i) = O(\rho)$. Finite-flow nonexpansion preserves this ambiguity at all later times.

Theorem 6.2. *For the class (38) there is a preparation time $s_*(\tau) \to \infty$, uniform in all length ratios and unitary parameters, such that $s_*/L_* \to 0$. Let $\widehat{K}_P(v)$ be the reconstruction of Theorem 5.2 for the finite flow starting at the selected tuple P. Then*

$$\eta_* = \int \sigma(H(s_*), \widehat{K}_P(0)) \to 0$$

uniformly over the stated initial class. Consequently (28) *holds with* $H(s_*+v)$, $\widehat{K}_P(v)$ *and initial error* η_*, *for* $0 \le v \le TL_*$. *Under* (34) *the all-time estimate of Theorem 5.3 holds after the same preparation.*

Proof. Fixed-base parabolic estimates for the component flows with bounded $C^{6,\beta}$ initial profiles give nondecreasing finite constants $C_n \ge 1$ controlling the required space and time derivatives on $0 \le s \le n$. They depend on R_0 and the fixed components, not on τ, U. Transfer $G_i(s,z)$ to the component exteriors. On each node join its two point values $G_i(s,p_{ie})$ and $U_{ij}G_j(s,p_{je})U_{ij}^*$ by the geodesic with parameter $-\varepsilon_e \log|u_e|$. The node curvature is zero, as in (32). On the i collar use the spatially constant reference $G_i(s,p_{ie})$ for both relative logarithms. On $\rho \le |u_e| \le 2\rho$, the difference between the component and node logarithms, with its first two derivatives, is at most

$$C_n(\rho + \varepsilon_e|\log\rho|)\rho^{-k}, \qquad k = 0,1,2.$$

The $O(\rho)$ term is the spatial variation of G_i from its nodal value, and the other term is the small node parameter. Use the same logarithmic cutoffs and curvature-difference estimate as in (33). The integrated collar residual is at most $C_n(\rho + \varepsilon_*|\log\rho|)$. The middle-node time residual is $O(C_n\rho^2)$ and the geometric transfer error is $O(C_n b\rho^{-N})$. Thus the resulting global preparation metric K_{pre} obeys

$$\sup_{0\le s\le n}\int |\mathcal{R}(K_{\mathrm{pre}})| \le C_n\delta_*, \qquad \delta_* = \frac{1+\log L_*}{L_*} + e^{-L_*}L_*^N \to 0.$$

The initial metric and $K_{\mathrm{pre}}(0)$ agree outside regions of area $O(\rho^2)$ and have common positive bounds. By (25) their integrated squared distance, and their integrated trace discrepancy, are bounded by $C\rho^2 + nC_n\delta_*$ up to time n.

For all sufficiently small parameters choose

$$s_* = \max\{n \in \mathbb{N} : nC_n \le \delta_*^{-1/2}\}.$$

It tends to infinity because each C_n is finite, and it is finite because $C_n \ge 1$. Also $s_* \le \delta_*^{-1/2}$, which implies $s_*/L_* \to 0$. At that time the component flows are exponentially close to P_i. The reconstruction with initial tuple P differs from the constant component metrics by the Green corrections; their squared integrals are $O(L_*^{-2})$. The remaining node regions have area $O(\rho^2)$. On the common matrix range, σ is comparable to squared distance and obeys a triangle bound up to a fixed constant. We obtain

$$\eta_* \le C_{R_0}\left(\sqrt{\delta_*} + e^{-\gamma_{R_0}(s_*-1)} + L_*^{-2}\right) \to 0.$$

Restart Theorem 5.2 at s_*, and Theorem 5.3 when its additional rate hypothesis holds. □

For physical times after preparation one may align the finite trajectory with time zero. Its speed is $O(L_*^{-1})$, and the uniform L^2 Lipschitz reconstruction gives, for $v \ge 0$,

$$\int d(H(s_*+v), \widehat{K}_P(s_*+v))^2 \le 2\int d(H(s_*+v), \widehat{K}_P(v))^2 + C(s_*/L_*)^2.$$

Thus the time shift costs a fixed factor in the preceding squared-error estimate and a vanishing additional term. Uniform tracking does not by itself assert convergence of a family of selected phases. To pass to a fixed positive rescaled-time limit, one must also assume convergence of those phases and of the normalized finite parameters.

7. Component holonomy and neck degeneration

We next allow the component connections to vary. The compact family in this section has one base parameter and four necks of length $L+O(1)$. The bundle parameters are independent of L. The conclusions therefore concern arbitrary component-to-neck scale ratios; they do not combine varying component holonomy with the six independent node parameters of Section 2.

7.1. **A compact family and its flat bundles.** Let E be the projective elliptic curve $y^2=(x-1)(x-2)(x-3)$, with origin o at infinity. The sections $s_0=x$ and $s_1=1$ of $\mathcal{O}_E(2o)$ have disjoint zero divisors $p+q$ and $2o$. In $E\times\mathbb{P}^1\times\Delta$ set

$$\mathcal{C}=\{s_0uv+ts_1(u^2+v^2)=0\},\qquad \pi:\mathcal{C}\longrightarrow\Delta. \tag{39}$$

Give E a fixed flat Kähler metric and $\mathbb{P}^1$ a fixed smooth Kähler metric. Restrict their product, with a positive disc term, to $\mathcal{C}$.

Proposition 7.1. *After shrinking Δ, (39) is a proper projective semistable family with smooth total surface and connected genus-three smooth fibres. Its central components are two copies E_+,E_- of E and two rational curves R_p,R_q, with dual graph $K_{2,2}$.*

Fix trace-free Hermitian matrices T_i and nonzero real harmonic one-forms ω_i on E_i, for $i=+,-$. The component connections

$$D_i=d-i\alpha_iT_i\omega_i,\qquad \alpha_i\in\mathbb{R},$$

the trivial connections on R_p,R_q, and arbitrary node identifications $U_{ij}\in\mathrm{SU}(r)$ define flat determinant-one bundles on a neighborhood of the whole central fibre. For every fixed α they are holomorphic bundles over the restricted family, with parallel metric h_0. Dependence on α is smooth and real.

Proof. At a node the equation is $xw+t(1+w^2)=0$. Its derivative in t is nonzero there. Elsewhere on the central fibre a spatial derivative is nonzero, including above o: divide by s_0 and use that s_1/s_0 has a double zero. Compactness gives a smooth total surface after restriction. The projection is proper and projective. Its fibres are effective Cartier divisors, and the local map to the disc is flat. The holomorphic Morse normal form gives $ab=t$ at each node.

For $t\neq0$, projection to E has degree two. The discriminant in the affine chart is x^2-4t^2, with four distinct simple zeros for small t. At o the equation $w+t(s_1/s_0)(1+w^2)=0$ and its reciprocal chart give two unramified points. Thus C_t is smooth. A disconnected double cover would be unramified, so it is connected. Riemann–Hurwitz gives genus three.

Use disjoint node bidiscs and component exteriors as in Proposition 2.3. Trivialize each exterior smoothly over the disc, with the lifts agreeing with the branch coordinate on its collars. Pull back ω_i from the fixed punctured component. It is closed and has zero puncture periods. On an incidence collar write $\omega_i=dF_{ie}$, with the central primitive normalized to vanish at the node. The frame change $G_{ie}=\exp(i\alpha_iT_iF_{ie})$ gives a parallel unitary frame. Give each node bidisc the trivial flat bundle and identify these parallel frames by the prescribed constant unitary maps. Only incidence overlaps occur; there are no triple overlaps.

These are flat isomorphisms, so the connections glue to a unitary flat connection with parallel determinant. Its $(0,1)$ part is integrable. The exterior smooth frames need not be holomorphic; the local parallel frames are. Properness permits a smaller disc on which this neighborhood contains every whole fibre. This proves the claimed total-family construction. □

Let A_i be the four limiting areas and put $\kappa_i = \int_{E_i} |\omega_i|^2$. The elliptic limiting metrics are flat, so $|\omega_i|^2 = \kappa_i/A_i$ is constant. All necks have leading capacity $2\pi/L$, where $L = \log(1/|t|)$.

The distinction between node identifications and based holonomy matters here. Killing the annular core loops gives a surjection

$$\pi_1(C_t) \longrightarrow \pi_1(E_+) * \pi_1(E_-) * \mathbb{Z}.$$

Fix paths c_{ie} from component basepoints to the nodes and put

$$\mathcal{U}_{ie} = \exp\left(i\alpha_i T_i \int_{c_{ie}} \omega_i \right), \qquad K_{ij} = \mathcal{U}_{ie}^{-1} U_{ij} \mathcal{U}_{je}.$$

On rational components $\mathcal{U}_{ie} = I$. Spanning-tree products of the K_{ij} identify the component basepoint fibres, and their cycle product represents the chosen based graph loop. A bare product of the U_{ij} need not do so. Changing paths changes the presentation of the same local system.

For all sufficiently small α_i, the parallel trace-free Hermitian endomorphisms are exactly the tuples satisfying

$$X_i = \mathrm{Ad}(U_{ij}) X_j, \qquad \alpha_i [T_i, X_i] = 0 \quad (i = +, -). \tag{40}$$

Indeed, choose a nonzero period of ω_i. For small nonzero α_i, exponentiation does not identify distinct eigenvalues of T_i at this period. Its holonomy commutant is therefore the commutant of T_i. Such an endomorphism is constant in the harmonic component gauge and unchanged by each G_{ie}; node compatibility is precisely (40). At $\alpha_i = 0$ the component condition is vacuous. The unitary correspondence [NS65] gives polystability, and stability exactly when this trace-free commutant is zero.

For example, take $r = 2$, $T_+ = \sigma_z$, $\alpha_- = 0$, and a harmonic form on E_+ with periods $(1, 0)$. Set the tree maps to I and the remaining node map to $W = (I + i\sigma_x)/\sqrt{2}$. Tree compatibility makes all $X_i = X$. When $\alpha_+ \neq 0$ is small the component condition gives $X = c\sigma_z$, and $W\sigma_z W^* = \sigma_y$ forces $c = 0$. The bundle is stable. At $\alpha_+ = 0$ the representation is generated by W and is reducible. This is actual stable-to-polystable variation, independent of t. It does not produce an unstable locus or a topologically nontrivial smooth $\mathrm{SU}(r)$ bundle on a curve.

7.2. **The coupled relative spectrum.** On $\mathcal{V} = \bigoplus_{i=1}^{4} \mathrm{Herm}_0(r)$ define

$$\begin{aligned} g_\alpha(X) &= \sum_{i=+,-} \alpha_i^2 \kappa_i \|[T_i, X_i]\|_{\mathrm{HS}}^2, \\ q_U(X) &= \sum_{e=ij} \|X_i - \mathrm{Ad}(U_{ij}) X_j\|_{\mathrm{HS}}^2, \\ Q_{t,\alpha,U}(X) &= g_\alpha(X) + \frac{2\pi}{L} q_U(X), \qquad M(X) = \sum_i A_i \,\mathrm{tr}(X_i^2). \end{aligned} \tag{41}$$

Write $\alpha_* = \max_i |\alpha_i|$, $m = 4(r^2 - 1)$, and let $\nu_0, \dots, \nu_{m-1}$ be the generalized eigenvalues of (Q, M), including zeros.

Theorem 7.2. *Let λ_k be the eigenvalues of D^*D on the real trace-free Hermitian endomorphisms of the bundle in Proposition 7.1. As $(|t|, \alpha_*) \to (0, 0)$,*

$$(1 - \epsilon)\nu_k \leq \lambda_k \leq (1 + \epsilon)\nu_k \quad (0 \leq k < m), \qquad \epsilon \longrightarrow 0,$$

uniformly in all U. There is a uniform complementary gap $\lambda_m \geq \gamma > 0$ in a sufficiently small product neighborhood. The exact kernel is (40). No restriction is imposed on $\alpha_i^2 L$ or on the distance to a reducible representation.

Proof. We give the component and matching estimates that replace those of Section 3. On a closed elliptic component a constant matrix has energy $\alpha_i^2\kappa_i\|[T_i, X_i]\|^2$. In flat coordinates the constant subspace is invariant under $D_i^* D_i$. After diagonalizing T_i, entry ab on a dual-lattice Fourier mode k has eigenvalue $|2\pi k - \alpha_i(\lambda_a(T_i) - \lambda_b(T_i))\omega_i|^2$. The nonzero Fourier modes have a uniform gap for small α_i.

Fix a small radius $a > 0$. A local primitive F satisfies $F(0) = 0$, $|F| \leq Ca$ and $|dF| \leq C$. On $a < |z| < 2a$ join X_i to the parallel section $\mathrm{Ad}(e^{i\alpha_i T_i F})X_i$ with a radial cutoff. Since

$$\|\,\mathrm{Ad}(e^{i\alpha_i T_i F})X_i - X_i\| \leq |\alpha_i F|\,\|[T_i, X_i]\|,$$

this modification changes the component energy by at most $Ca^2\alpha_i^2\|[T_i, X_i]\|^2$. In the flat node frames interpolate affinely in log radius over $\ell_a = L + 2\log a$. For the resulting trial section JX,

$$E(JX) \leq (1 + Ca^2 + o_a(t))g_\alpha(X) + \frac{2\pi}{\ell_a} q_U(X),$$
$$|\|JX\|^2 - M(X)| \leq (Ca^2 + o_a(t))M(X).$$

Metric transfer errors are multiplicative because the pulled-back connection is fixed. These bounds give the relative upper estimate and $\lambda_{m-1} \leq C(\alpha_*^2 + L^{-1})$.

For the lower estimate, let Ω_a be a punctured elliptic component and let ω_a be the absolute harmonic representative of the class of $\omega|_{\Omega_a}$. Thus ω_a is closed, coclosed, has zero normal component, and $\omega = \omega_a + df_a$. Write $\kappa_a = \|\omega_a\|^2$. Then $\kappa_a \to \kappa$ as $a \to 0$. Indeed ω is a competitor, so $\kappa_a \leq \kappa$. A bounded-energy weak limit retains its periods and is harmonic away from the punctures. An isolated singularity of an L^2 harmonic one-form on a surface is removable: even a simple pole in its holomorphic Laurent expansion has infinite L^2 norm. The limit is therefore ω, and lower semicontinuity proves the assertion.

Gauge by $e^{i\alpha T f_a}$. For an entry with eigenvalue difference δ of T, the connection becomes $d - i\beta\omega_a$, where $\beta = \alpha\delta$. Write $v = z + w$, with $\int_{\Omega_a} w = 0$. The absolute boundary condition gives $\langle dw, \omega_a z\rangle = 0$. For small $|\beta|$, Poincaré implies $\|w\| \leq C_a\|(d - i\beta\omega_a)w\|$. The remaining cross term is bounded by $C_a\beta^2|z|\|w\|$. Completing the square therefore gives

$$\|(d - i\beta\omega_a)v\|^2 \geq \tfrac{1}{2}\|(d - i\beta\omega_a)w\|^2 + (\beta^2\kappa_a - C_a\beta^4)|z|^2,$$
$$\|w\|_{H^1}^2 + \beta^2|z|^2 \leq C_a\|(d - i\beta\omega_a)v\|^2.$$

These inequalities remain valid at $\beta = 0$. Summing entries defines component means X_i and gives, with multiplicative $1 + o_a(t)$ changes on the nearby fibre,

$$E_i \geq (1 - C_a\alpha_*^2)\alpha_i^2\kappa_{i,a}\|[T_i, X_i]\|^2,$$
$$\|w_i\|_{H^1}^2 + \alpha_i^2\|[T_i, X_i]\|^2 \leq C_a E_i.$$

On rational components take the ordinary mean.

The mean gauge and the parallel node frame differ by $e^{i\alpha_i T_i(f_{i,a} - F_{ie})}$. On a fixed collar its action on X_i differs from X_i by at most $C_a|\alpha_i|\|[T_i, X_i]\|$. The trace bound on w_i thus gives $|c_{ie} - X_i| \leq C_a E_i^{1/2}$ for the parallel-frame boundary averages. In particular there is no additive error $O(\alpha_i)\|X_i\|$. Separate the disjoint component and neck energies as E_{c} and E_{n}. Angular averaging on the annuli yields

$$q_U(X)^{1/2} \leq \sqrt{\ell_a/(2\pi)}\, E_{\mathrm{n}}^{1/2} + C_a E_{\mathrm{c}}^{1/2}.$$

After multiplication by $\sqrt{2\pi/L}$ and Young's inequality, this gives

$$\frac{2\pi}{L} q_U(X) \le (1 + O_a(L^{-1/2}))E_{\mathrm{n}} + O_a(L^{-1/2})E_{\mathrm{c}}.$$

Combining it with the component estimate, without counting either energy twice, gives

$$Q(X(s)) \le \big(1 + \theta(a) + C_a\alpha_*^2 + O_a(L^{-1/2}) + o_a(t)\big)E(s), \qquad \theta(a) \to 0. \tag{42}$$

We also need the mass comparison. For a normalized eigensection of eigenvalue $\lambda \le 1$, the preceding bounds give $\|w_i\|_{H^1} \le C_a\sqrt{\lambda}$ and bounded means. In the mean gauge the central operator applied to X_i is $\alpha_i^2|\omega_{i,a}|^2 \operatorname{ad}(T_i)^2 X_i$, bounded by $C_a|\alpha_i|\sqrt{E_i}$. The metric perturbation adds at most $o_a(t)|\alpha_i|\|[T_i, X_i]\|$. Interior $W^{2,2}$ estimates on fixed enlarged collars therefore control the boundary values in C^0. The enlargement passes slightly into the flat node chart and introduces no shrinking-domain estimate. Harmonic extension in that chart, followed by Lemma 3.2 for its zero-boundary remainder, rules out annular mass loss. For a combination of at most $m+1$ eigenvectors of eigenvalues at most $\Lambda \le 1$, the result is

$$|M(X(s)) - \|s\|^2| \le (Ca^2 + C_a\sqrt{\Lambda} + o_a(t))\|s\|^2.$$

This is the mass argument of Lemma 3.3 with the energy-controlled component gauges above.

If λ_m tended to zero, these means would inject an $(m+1)$-dimensional space into $\mathcal{V}$, a contradiction. The same mass estimate and (42), applied to the first $k+1$ eigensections, give the relative lower estimate by min–max. Choose a first and then choose $|t|$ and α_* independently small. This proves uniformity for every ratio $\alpha_i^2 L$. Finally $Q = 0$ is exactly (40), already identified with the actual parallel sections. □

7.3. **The exact component field.** For $H \in \mathcal{P}_r$ set $f_T(H) = \operatorname{tr}(TH^{-1}TH) - \operatorname{tr}(T^2)$. In the smooth harmonic component frame a spatially constant metric has

$$H^{-1}\partial_{D_i}H = -i\alpha_i(H^{-1}T_iH - T_i)\omega_i^{1,0}, \qquad F_H = \alpha_i^2[T_i, H^{-1}T_iH]\,\omega_i^{1,0} \wedge \omega_i^{0,1}.$$

Since $2i\Lambda(\omega_i^{1,0} \wedge \omega_i^{0,1}) = |\omega_i|^2$, its exact component HYM equation is

$$H^{-1}\dot{H} = -\alpha_i^2|\omega_i|^2[T_i, H^{-1}T_iH].$$

Differentiation gives $df_T(HS) = \operatorname{tr}([T, H^{-1}TH]S)$. The gradient is H-self-adjoint and trace free. It is geodesically convex: along $H(u) = H_0^{1/2}e^{uK}H_0^{1/2}$, put $B = H_0^{-1/2}TH_0^{1/2}$ and diagonalize K with eigenvalues k_a. Then

$$\operatorname{tr}(TH(u)^{-1}TH(u)) = \sum_{a,b} |B_{ba}|^2 e^{u(k_a - k_b)}.$$

For $H = e^S$, the Hermitian symmetry of T also gives

$$f_T(e^S) = 2\sum_{a<b} |T_{ab}|^2(\cosh(s_a - s_b) - 1) \ge \tfrac{1}{2}\|[T, S]\|^2, \qquad f_T(e^{uS}) \le u^2 f_T(e^S) \quad (0 \le u \le 1).$$

Equip $\mathcal{P}_r^4$ with the masses A_i and define

$$\Phi_{t,\alpha,U}(H) = \sum_{i=+,-} \alpha_i^2\kappa_i f_{T_i}(H_i) + \frac{\pi}{L}\sum_{e=ij} d(H_i, U_{ij}H_jU_{ij}^*)^2.$$

Its negative gradient is

$$A_iH_i^{-1}\dot{H}_i = -\alpha_i^2\kappa_i[T_i, H_i^{-1}T_iH_i] + \frac{2\pi}{L}\sum_{j:ij} B_{ij}, \qquad B_{ij} = \log(H_i^{-1}U_{ij}H_jU_{ij}^*), \tag{43}$$

with the component term omitted on rational vertices. All derivatives use physical time.

Convexity gives nonexpansion of this flow. Radial contraction towards the identity gives $\Phi_{t,\alpha,U}(e^{uS}) \le u^2\Phi_{t,\alpha,U}(e^S)$, and the distance-decreasing logarithm gives $\Phi_{t,\alpha,U}(e^S) \ge Q(S)/2$. Consequently

$$R(s) := \tfrac{1}{2}M(\log H(s)) \le R(0)e^{-2\nu_0 s}. \tag{44}$$

Indeed $R' \le -2\Phi_{t,\alpha,U} \le -2\nu_0 R$. This also confines every trajectory to its initial product ball and proves all-time existence, including $\nu_0 = 0$. On that ball its speed, and the physical time derivative of any fixed smooth tuple function, are bounded by $C\zeta$, where $\zeta = \alpha_*^2 + L^{-1}$. The speed in neck time need not be bounded.

7.4. **Actual coupled tracking.** Put

$$e(t,\alpha) = \alpha_* + \frac{(1+\log L)^2}{L}.$$

The base, component and gluing parameters remain fixed during each trajectory.

Theorem 7.3. *For $|t|$ and α_* independently sufficiently small, and initial tuples in a fixed bounded ball, the finite flow (43) defines a smooth determinant-one comparison metric $\widehat{K}(s)$ on C_t, for all $s \ge 0$, with uniform positive bounds and*

$$\sup_{s\ge 0}\int_{C_t} |\mathcal{R}(\widehat{K}(s))|_{\mathrm{op}} \le C\left(\frac{\alpha_*}{L} + \frac{(1+\log L)^2}{L^2}\right). \tag{45}$$

On each fixed compact component region, in the harmonic gauge, $d(\widehat{K}(s), H_i(s)) \le C_K/L$. For an actual HYM flow with fixed positive initial bounds, let $\eta = \int_{C_t}\sigma(H(0),\widehat{K}(0))$. Then for every $T \ge 0$,

$$\sup_{0\le s\le TL}\int_{C_t} d(H(s),\widehat{K}(s))^2 \le \eta + CTe(t,\alpha). \tag{46}$$

All constants are independent of T, U and the ratios $\alpha_i^2 L$. If in addition $L\nu_0 \ge \nu_* > 0$, then

$$\sup_{s\ge 0}\int_{C_t} d(H(s),\widehat{K}(s))^2 \le \eta + C_{\nu_*}e(t,\alpha)(1+|\log e(t,\alpha)|) \tag{47}$$

for small $e(t,\alpha)$.

Proof. Use the component Green functions of Section 5 and set

$$\xi_i = -\sum_{e=ij} B_{ij}g_{ie}, \qquad V_i = \frac{2\pi}{A_i}\sum_{j:ij} B_{ij}, \qquad K_i = H_i e^{\varepsilon\xi_i}, \quad \varepsilon = L^{-1}.$$

Off the punctures $\Delta_i\xi_i = -V_i$. On a component cut at radius $\rho/2$, the fixed Green estimates give

$$\|\xi_i\|_\infty \le C(1+|\log\rho|), \qquad |\nabla^k\xi_i| \le C(1+|u|^{-k}) \quad (k=1,2),$$
$$\int(|\xi_i| + |\xi_i|^2 + |\nabla\xi_i|) \le C, \qquad \int|\nabla\xi_i|^2 \le C(1+|\log\rho|),$$
$$\int|\xi_i|\,|\Delta_i\xi_i| \le C, \qquad \int|\dot\xi_i| \le C\zeta.$$

The derivative bound is written near each node in its coordinate u; away from the nodes it is uniform. The last estimate uses the physical speed bound; no bound on $L\dot H_i$ is used.

In the harmonic gauge write $h = \varepsilon\xi_i$ and $C_H = H^{-1}\partial_{D_i}H$. The exact identity

$$K^{-1}\partial_{D_i}K = e^{-h}C_He^h + e^{-h}\partial_{D_i}e^h$$

and the integral formula for the derivative of the exponential imply, when $|h| \le 1$,

$$|2i\Lambda F_K - 2i\Lambda F_H - \Delta_i h| \le C\big(|\alpha_i||\nabla h| + \alpha_i^2|h| + |\nabla h|^2 + |h||\Delta_i h|\big).$$

Here $|C_H| = O(|\alpha_i|)$ and its covariant derivative is $O(\alpha_i^2)$. The terms linear in h contain either a connection coefficient and a derivative of h, or two connection coefficients and h. Differentiating the exponential produces the two quadratic terms. The time identity $K^{-1}\dot K = e^{-h}(H^{-1}\dot H)e^h + e^{-h}\partial_s e^h$ has error $C(|\dot h| + \zeta|h|)$. Since $|\omega_i|^2 = \kappa_i/A_i$, (43) cancels the leading terms exactly:

$$H_i^{-1}\dot H_i + 2i\Lambda_i F_{H_i} + \varepsilon\Delta_i\xi_i = 0.$$

Thus the integrated component residual is at most

$$C\{\varepsilon(\zeta + \alpha_* + \alpha_*^2) + \varepsilon^2(1 + |\log\rho|)\}.$$

There is no uncancelled pure component error of order α_*^4. Transfer to the smoothing adds $C|t|\rho^{-N}$ for a fixed N.

At incidence ie write $\xi_i = -B_{ij}\log|u| + c_{ie} + q_{ie}(u)$, where $q_{ie}(0) = 0$, and put $P_{ie} = H_i e^{\varepsilon c_{ie}}$. In the node parallel frame use

$$N_e = \mathrm{Geo}(P_{ie}, U_{ij}P_{je}U_{ij}^*; -\varepsilon\log|u|).$$

Geodesic reversal and $uv = t$ give exact compatibility from both ends. This node metric has zero spatial Chern curvature and physical time velocity $O(\zeta)$, so its middle-node residual has integral $O(\zeta\rho^2)$.

The component metric in this parallel frame is $K_i^{\mathrm{par}} = G_{ie}^* K_i G_{ie}$, where $G_{ie} = e^{i\alpha_i T_i F_{ie}}$ and $F_{ie}(0) = 0$. On the incidence annulus, in the logarithm $\ell_i(K) = \log(H_i^{-1}K)$, the endpoint expansion used in (33) gives

$$|D^k(\ell_i(K_i^{\mathrm{par}}) - \ell_i(N_e))| \le C\{\alpha_*\rho^{1-k} + \varepsilon\rho^{1-k} + \varepsilon^2(1 + |\log\rho|)\rho^{-k}\}, \quad k = 0, 1, 2.$$

The new first term comes from $G_{ie} - I$. Each estimate differentiated in physical time gains a factor $C\zeta$. Interpolate the two H_i-self-adjoint, trace-free logarithms and exponentiate. The curvature-difference estimate of Section 5, with $|Dn| \le C\varepsilon/\rho$ and $|D^2 n| \le C\varepsilon/\rho^2$ for $n = \ell_i(N_e)$, gives the integrated collar bound

$$C\{\alpha_*\rho + \varepsilon\rho + \varepsilon^2(1 + |\log\rho|)^2 + \alpha_*^2\rho^2 + \zeta\rho^2\}.$$

Choose $\rho = L^{-1}$. Then $\varepsilon|\log\rho| \to 0$, the exponentially small transfer error is absorbed, and all pieces give (45). They remain in a common positive matrix range for all physical time. On fixed component regions the Green correction is $O(L^{-1})$.

Lemma 5.1 is local in parallel holomorphic frames and applies to this flat bundle. Integrating (45) proves (46), without a factor exponential in $\alpha_*^2 s$.

For the all-time statement, Theorem 7.2 gives $\lambda_0 \ge \nu_*/(2L)$. The actual trace coercivity used in Theorem 5.3 remains valid in the local parallel frames: for $H = e^S$ and $cI \le H \le CI$,

$$\frac{c}{2}\|S\|_{L^2}^2 \le F := \int \mathrm{tr}(H - I) \le \frac{C}{2}\|S\|_{L^2}^2, \qquad -F' \ge \frac{c^2}{C}\int |DS|^2.$$

It implies exponential decay in s/L with rate bounded below by a multiple of ν_*. Equation (44) gives the same type of bound for the finite flow. All correction coefficients vanish at the identity tuple, which is fixed by the component gauge changes. The reconstructed metric consequently has an exponential tail as well. Combine these tails beyond time TL with (46) before that time and take T proportional to $|\log e(t, \alpha)|$. This proves (47). □

The additional rate condition is nonempty even when $\alpha_*^2 L \to \infty$. For the rank-two example above write $Q = hq + aP$, where $h = 2\pi/L$, $a = \alpha_+^2\kappa_+$ and $P(X) = \|[\sigma_z, X_+]\|^2$. The kernel computation gives $\ker q \cap \ker P = 0$. If $a/h \geq \eta_0 > 0$, then $Q \geq h(q + \eta_0 P)$ and $L\nu_0$ has a positive lower bound. As $a/h \to 0$ that bound fails; the bounded-neck-time theorem still applies.

7.5. **The initial layer with varying component holonomy.** Use the norm (38) with $\rho = L^{-1}$, the four components of (39), and the harmonic component frames. Denote it by $\mathcal{N}_{t,\alpha,U}$. It again defines a fixed-radius nonempty open class $\mathcal{N}_{t,\alpha,U}(S) < R_0$ on each smooth fibre, with only a C^0 bound imposed in the middle necks. For admissible component extensions take the phases P_i of their *trivial* component flows from Theorem 6.1.

Theorem 7.4. *In the joint limit $(|t|, \alpha_*) \to (0,0)$, there is a preparation time $s_*(t,\alpha;R_0) \to \infty$ such that $(\alpha_*^2 + L^{-1})s_* \to 0$ and*

$$\eta_* = \int_{C_t} \sigma(H(s_*), \widehat{K}_P(0)) \longrightarrow 0$$

uniformly over this initial class and all unitary parameters. The estimates of Theorem 7.3 apply after s_* with initial error η_*. The finite flow may instead be aligned with physical time zero at the cost of a fixed factor in the squared error and an additional $O((\zeta s_*)^2)$.

Proof. The preparation construction in Theorem 6.2 uses trivial component solutions $G_i(s,z)$ with uniform derivative bounds C_n on $0 \leq s \leq n$. Such bounds follow on each fixed component by compactness of the bounded $C^{6,\beta}$ initial class in $C^{4,\beta/2}$, continuous dependence, and finite-time parabolic regularity. Enlarge $C_n \geq 1$ to be nondecreasing.

In a harmonic frame, replacing ordinary ∂ by ∂_{D_i} introduces first-order errors $O(C_n|\alpha_i|)$ and zero-order errors $O(C_n\alpha_i^2)$. In the node frame the additional gauge factor is $I + O(\alpha_*\rho)$. Use the spatially constant nodal value $G_i(s, p_{ie})$ as reference for both collar logarithms, as in Theorem 6.2. Their derivative difference is bounded by $C_n(\rho + L^{-1}|\log\rho|)\rho^{-k}$ for $k = 0,1,2$. The middle-neck time error is $O(C_n\rho^2)$ and transfer costs $C_n|t|\rho^{-N}$. Thus the global preparation metric satisfies

$$\sup_{0\leq s\leq n} \int |\mathcal{R}(K_{\mathrm{pre}})| \leq C_n\delta, \qquad \delta = \alpha_* + \frac{1+\log L}{L} + |t|L^N \to 0.$$

Its initial discrepancy has integral $O(\rho^2)$ because the metrics agree on the exteriors. Initial neck derivatives do not enter this estimate.

Choose $s_* = \max\{n : nC_n \leq \delta^{-1/2}\}$. Then $s_* \to \infty$, $s_*C_{s_*}\delta \leq \sqrt{\delta}$ and $\zeta s_* \leq C\sqrt{\delta} \to 0$. The trace comparison, fixed-component phase convergence, and the bounded L^2 norms of the Green functions give

$$\eta_* \leq C_{R_0}\{\sqrt{\delta} + e^{-\gamma_{R_0}(s_*-1)} + L^{-2}\}.$$

Restart the coupled tracking theorem.

Two admissible extensions differ only in discs of area $O(\rho^2)$. The phase contraction estimate gives an $O(\rho)$ ambiguity in P. Nonexpansion of (43) preserves this bound for all time. The reconstruction is uniformly Lipschitz from the confined tuple ball to spatial L^2: its Green terms have uniformly bounded L^2 coefficients, and node geodesics and collar interpolations have bounded endpoint derivatives. The same observation and the physical speed bound give

$$\int d(H(s_*+v), \widehat{K}_P(s_*+v))^2 \leq 2\int d(H(s_*+v), \widehat{K}_P(v))^2 + C(\zeta s_*)^2.$$

This proves the time alignment and extension-independence assertions. □

The phase theorem here uses that the limiting component connections become trivial as $\alpha \to 0$. It does not select phases for arbitrary nontrivial limiting local systems. Uniform all-time comparison retains $L\nu_0 \geq \nu_* > 0$; it has not been deduced at a degenerating smallest rate from the relative spectrum alone.

8. The Hodge metric on the gluing locus

Return to the component-trivial bundles of Section 2, and restrict the quartic family to the one-parameter slice $\tau_e = t$ for every edge. Thus $L = \log(1/|t|)$ is common to all six necks. We compare the actual harmonic-one-form L^2 metric on a locus of flat-bundle moduli with a finite quotient metric. The mass weights in (7) do not enter this one-form comparison.

8.1. **The graph quotient and its embedding.** Write $\Gamma = K_4$, $d = r^2-1$ and $g_\Gamma = b_1(\Gamma) = 3$. The irreducible graph representation space is

$$\mathcal{M}_\Gamma^{\mathrm{irr}} = \mathrm{SU}(r)^{E,\mathrm{irr}} / \mathrm{SU}(r)^V, \qquad U_{ij} \longmapsto g_i U_{ij} g_j^{-1}.$$

After dividing the vertex gauge group by its ineffective diagonal center, this is a free proper action. The quotient is a smooth real manifold of dimension $(|E| - |V|)d = (g_\Gamma - 1)d = 2d$. Choose an orientation for each edge. A tangent representative is $a_e = \dot{U}_e U_e^{-1} \in \mathfrak{su}(r)$. Put

$$(\delta_U x)_{ij} = x_i - \mathrm{Ad}(U_{ij}) x_j, \qquad G_{\Gamma,U}([a]) = \min_{x \in \mathfrak{su}(r)^V} \sum_e \|a_e + (\delta_U x)_e\|^2. \tag{48}$$

Here $\langle X, Y\rangle = -\operatorname{tr}(XY)$. This is the quotient metric of the bi-invariant product metric on the edges. The notation denotes its squared norm on a tangent class.

Let $\mathcal{M}_t^{\mathrm{irr}}$ be the moduli space of irreducible flat $\mathrm{SU}(r)$ connections on C_t. Its tangent space at D is $H^1(C_t; \operatorname{ad} D)$. Irreducibility gives $H^0 = 0$ and duality gives $H^2 = 0$, so the local quotient is smooth. We use the Hodge metric: a class has the squared L^2 norm of its unique D-harmonic one-form representative, equivalently the minimum of $\int |b|^2$ over its closed representatives modulo exact forms. These deformation and pairing conventions are those of [Gol84, Section 1]. Via [NS65] this is a metric on stable rank-r degree-zero bundles with trivial determinant.

Proposition 2.3 gives a map $\Phi_t : \mathcal{M}_\Gamma^{\mathrm{irr}} \to \mathcal{M}_t^{\mathrm{irr}}$ by inflation along $\pi_1(C_t) \twoheadrightarrow \pi_1(\Gamma)$. It is an embedding. Surjectivity of this group map shows that conjugate inflated representations were already conjugate. Evaluation on fixed lifts of the free graph generators gives a continuous left inverse on the image, smooth in local representation slices. The differential is injective: an inflated cocycle that is a coboundary on the surface has the same coboundary expression on these lifts. This proves both immersion and the topology of the embedding. Denote the pullback Hodge metric by G_t. The construction is a unitary Schottky locus; the free-group construction itself is classical [Flo01, Section 3, Proposition 1].

Theorem 8.1. *There is a constant C, depending only on the fixed family, rank and exterior collars, such that for large L,*

$$\left(1 - \frac{C}{L}\right) \frac{2\pi}{L} G_{\Gamma,U}([a]) \leq G_{t,U}([a]) \leq \left(1 + \frac{C}{L}\right) \frac{2\pi}{L} G_{\Gamma,U}([a]). \tag{49}$$

This holds for every irreducible U and tangent class, uniformly as U approaches reducibility. At a reducible U the same energy inequalities hold on $C^1(\Gamma; \operatorname{ad} U)/\operatorname{im} \delta_U$; no smooth moduli interpretation there is asserted.

For a connected relatively compact open subset $\Omega \subset \mathcal{M}_\Gamma^{\mathrm{irr}}$, let $d_{t,\Omega}$ and $d_{\Gamma,\Omega}$ be the intrinsic path distances using paths in Ω. Then

$$\sqrt{1 - C/L}\, d_{\Gamma,\Omega} \le \sqrt{\frac{L}{2\pi}}\, d_{t,\Omega} \le \sqrt{1 + C/L}\, d_{\Gamma,\Omega}. \tag{50}$$

The image of Φ_t is Lagrangian in the surface moduli space with its cohomological symplectic form.

8.2. **Affine primitives and the energy infimum.**

Proof. Fix a small collar radius a, independent of t. The thick components and the annuli of logarithmic length $\ell_a = L + 2\log a$ partition C_t. Each has a flat unitary frame.

An inflated graph tangent class restricts to zero cohomology on every thick piece and every annulus: the represented family has identically trivial holonomy on the fundamental group of each such patch. Thus any closed representative b has single-valued local primitives. In the fixed flat frames write $b_i = df_i$. On the left overlap choose the node primitive to satisfy $f_e = f_i$. On the right overlap it has the affine jump

$$f_e = \mathrm{Ad}(U_{ij}) f_j + a_e. \tag{51}$$

The constants represent the graph cochain class $[a]$. To check the sign, if varying local parallel frames obey $P_j(s) = P_i(s)U_{ij}(s)$, their infinitesimal changes satisfy $dk_i = -b_i$ and $k_i = \mathrm{Ad}(U_{ij})k_j - a_{ij}$. Taking $f_i = -k_i$ gives (51). Changes of primitive constants change the cochain by an element of $\operatorname{im}\delta_U$. Conversely these affine jumps construct every graph cochain class.

Choose a minimizer x in (48). Use constant component potentials $-x_i$, and interpolate in log radius on edge ij from $-x_i$ to $a_e - \mathrm{Ad}(U_{ij})x_j$. The difference of the endpoints is $a_e + (\delta_U x)_e$. Component derivatives vanish and one-form energy is conformally invariant in two dimensions. The resulting energy is exactly

$$\frac{2\pi}{\ell_a} \sum_e \|a_e + (\delta_U x)_e\|^2.$$

Piecewise affine primitives lie in H^1 and give weakly closed L^2 one-forms. Smoothing the profiles near their ends preserves the affine jumps and approximates the same energy. Equivalently one may use the weak Hodge projection. The variational characterization of G_t gives the upper estimate in (49).

The potentials x_i themselves can be large near a stabilizer change. Their size does not enter this bound: their component derivatives are zero and the annular derivatives depend only on the minimizing residual.

For the lower estimate take any smooth closed representative b of the class. Let x_i be the thick-component mean of f_i, and c_{ie} its boundary circle mean. Fixed-domain Poincaré and trace estimates give

$$\|c_{ie} - x_i\| \le C_a \|b\|_{L^2(\text{component } i)}.$$

The constants are uniform in t, U: the fixed thick metrics converge smoothly and the frames are unitary. Circle averaging and Cauchy–Schwarz along the log cylinder give

$$E_e(b) \ge \frac{2\pi}{\ell_a} \|a_e + \mathrm{Ad}(U_{ij})c_{je} - c_{ie}\|^2.$$

Writing E_{n} and E_{c} for the disjoint neck and component energies, we obtain

$$\begin{aligned} G_{\Gamma,U}([a])^{1/2} &\le \left(\sum_e \|a_e - (\delta_U x)_e\|^2 \right)^{1/2} \\ &\le \sqrt{\ell_a/(2\pi)}\, E_{\mathrm{n}}^{1/2} + C_a E_{\mathrm{c}}^{1/2} \\ &\le \sqrt{\ell_a/(2\pi) + C_a^2} \left(\int_{C_t} |b|^2 \right)^{1/2}. \end{aligned}$$

The last line uses $E_{\mathrm{n}} + E_{\mathrm{c}} = \int |b|^2$. Taking the infimum over representatives gives

$$G_{t,U}([a]) \ge \left(\frac{\ell_a}{2\pi} + C_a^2 \right)^{-1} G_{\Gamma,U}([a]),$$

which is the required lower estimate. One fixed collar radius suffices. There is no division by an eigenvalue of $\delta_U^* \delta_U$, and no zero-form mass estimate is needed. Fixed-fibre compact Hodge theory identifies the infimum with the actual harmonic norm. The same proof at a reducible parameter is an inequality on cohomology, independent of whether the moduli quotient is smooth.

Taking square roots of the pointwise quadratic inequalities and integrating along a path proves the corresponding length bounds. Taking infima over the same paths in Ω proves (50).

Finally each graph class has a smooth representative supported in the node annuli with only a radial component. Two such representatives wedge to zero on each annulus; different annuli are disjoint. The form $\int -\operatorname{tr}(b_1 \wedge b_2)$ therefore vanishes on the image of Φ_t. All normalization components are rational, so $g(C_t) = g_\Gamma = 3$. The image dimension $2d$ is half the surface-moduli dimension $4d$, proving the Lagrangian assertion. □

The Lagrangian interpretation concerns classical Schottky geometry. The quantitative assertion here is the relative comparison of the actual unweighted Hodge norm with the edge quotient norm and its coefficient $2\pi/L$. Daskalopoulos–Wentworth [DW93, Proposition 3.23 and Corollary 3.24] prove weighted operator continuity and kernel continuation in a degeneration, and obtain local moduli geometry. Their operator analysis allows arbitrary rank and retains boundary zero modes. It does not state (49); the argument above compares unweighted energy infima directly. In particular it does not invoke the limiting zero-cokernel hypothesis used for their kernel continuation.

The leading metric has no area weights because it minimizes one-form energy. The areas remain essential to the zero-form spectrum and finite HYM evolution. Equation (50) uses paths inside the indicated locus; it makes no assertion about shortcuts through the ambient moduli space or distances across reducible strata.

9. Marked observations and integral monodromy

We specify two inverse problems. The first recovers projective bundle transport on a known graph. The second recovers an unknown incidence graph and then the integral monodromy of the curve family. Both use component averages of low eigensections, in addition to eigenvalues. The integral conclusion also uses the known first-order smoothing and the orientation of the base loop. These observations are substantially richer than an unordered spectrum.

9.1. **The component-trivial construction on other graphs.**

We first record the geometric hypotheses needed to apply the preceding arguments beyond the quartic.

Proposition 9.1. *Let $\pi : \mathcal{Y} \to \Delta$ be a proper holomorphic nodal curve family with smooth total surface, connected smooth fibres, and a reduced central fibre with smooth normalization components Y_i. Suppose its finite dual graph is connected and has no self-edges, every node has coordinates $u_e v_e = t$, and the fibre metrics are induced by a fixed smooth Kähler form on the total surface. Use component-trivial flat bundles with constant $\mathrm{SU}(r)$ incidence maps on the exterior/node cover.*

Then the spectral, finite-flow, actual-flow and entry results of Sections 3–6 hold with $m = |V|(r^2-1)$, component areas A_i, and all weights equal to $2\pi/L$, $L = \log(1/|t|)$. Constants may depend on the fixed family and rank. If $b_1(\Gamma) \geq 2$, the metric and intrinsic-distance comparisons of Theorem 8.1 hold on the irreducible graph-gluing locus. That locus is Lagrangian when all Y_i are rational.

Proof. Choose disjoint node bidiscs and connected component exteriors. Their only overlaps are the two disjoint incidence collars at each node. Constant unitary maps therefore give the flat bundle and parallel determinant exactly as in Proposition 2.3; properness makes the cover contain the entire nearby fibres. Van Kampen gives the quotient onto $\pi_1(\Gamma)$ by killing the patch groups.

The required analytic estimates use only this cover and fixed component geometry. A node cut at radius a has area $O(a^2)$, logarithmic length $L + 2\log a$, and exact interpolation energy $2\pi\|X_i - \mathrm{Ad}(U_{ij})X_j\|^2/(L + 2\log a)$. The zero-boundary estimate of Lemma 3.2 is local in the smooth ambient node chart. Fixed-component Poincaré, trace and elliptic estimates give (12) and (13) with the new finite number of vertices. The same min–max and dimension argument gives exactly m weak modes and the gap.

The product matrix space is now $\mathcal{P}_r^{|V|}$. Its finite potential, convexity and coercivity are unchanged sums over this graph. On each fixed closed Y_i, the mean-zero Green functions have precisely the logarithmic singularity and integral estimates used in Section 5. The construction therefore gives the same integrated residual, trace comparison and all-time estimate under the normalized spectral lower bound. The fixed-component phase theorem holds on any of these compact curves. Its preparation construction uses only finitely many such flows, fixed-coordinate collars and the node area bound, so the same open initial class defined by the extension norm and the diagonal preparation time apply.

Finally, the proof of Theorem 8.1 uses the patch-trivial representation, affine primitives, fixed exterior trace inequalities and the common annular length. These hypotheses are all present. Its quotient dimension is $(b_1(\Gamma) - 1)(r^2 - 1)$. The radial representatives remain isotropic; if every component is rational, the smooth genus is $b_1(\Gamma)$ and the dimension is half that of the surface moduli space. □

9.2. **The measured operator.** For the quartic with independent node lengths, let $\mathcal{S}_{\tau,U}$ be the first m eigenspaces, counted with multiplicity and including the exact kernel. More precisely, this is the m-dimensional spectral subspace below the uniform complementary threshold. On a fixed exterior of component i choose a parallel unitary frame. Fix a small cutoff a_0 and define the averaging map

$$(T_{\tau,U}s)_i = \frac{1}{\mathrm{Area}(\Omega_i(a_0/2,\tau))} \int_{\Omega_i(a_0/2,\tau)} s_i \, dA_\tau.$$

Its target $\mathcal{V}$ carries the area inner product M. The observations are the first m eigenvalues and the columns of $T_{\tau,U}$ in an L^2-orthonormal eigenbasis, together with the known component labels, areas, local frames and normalized node parameters. No identification of component frames across a node is observed.

Lemma 9.2. *The map $T = T_{\tau,U}$ is invertible for small parameters, and $\|T^*T - I\| \to 0$ uniformly in U and all node-length ratios. Set*

$$G = TT^*, \qquad J = T^*G^{-1/2}, \qquad \widehat{B}_{\tau,U} = J^*(\Delta_{\tau,U}|_{\mathcal{S}_{\tau,U}})J.$$

Adjoints on $\mathcal{V}$ use M. Then J is an isometry and

$$\|\widehat{B}_{\tau,U} - B_{\tau,U}\|_M = o(w_*), \qquad w_* = \max_e w_e = \frac{2\pi}{L_*}. \tag{52}$$

All these matrices are determined by the declared observations and are independent of the chosen orthonormal eigenbasis. For the common-neck families of Proposition 9.1 the same assertion holds with $w_* = 2\pi/L$.

Proof. The energy of every unit vector in the low space is at most Cw_*. Let $b^{(a)}$ be the means at a smaller fixed radius $a < a_0/2$ used in Lemma 3.3. The two averaging regions have a common fixed positive-area subset. Poincaré on the larger truncated component gives

$$\|b^{(a)}(s) - Ts\|_M \le C_a\sqrt{w_*}\,\|s\|_{L^2}.$$

The low-space mass estimate consequently gives $|\|Ts\|_M^2 - \|s\|^2| \le (Ca^2 + o_a(1))\|s\|^2$. Let the parameters tend to zero and then $a \to 0$. This proves the uniform asymptotic isometry and invertibility. The polar formula gives $J^*J = I$ and $TJ = G^{1/2}$.

Apply (12) at the fixed radius a_0 to Jv. Writing $H = G^{1/2}$ and $\beta_\tau = (1 + C_{a_0}\sqrt{w_*})^{-2}$, we obtain the quadratic-form inequality

$$\widehat{B}_{\tau,U} \ge D_{\tau,U} := \beta_\tau H B_{\tau,U} H.$$

Since $\|B_{\tau,U}\|_M \le Cw_*$, we have $\|D_{\tau,U} - B_{\tau,U}\|_M = o(w_*)$. Relative spectral comparison gives $\operatorname{tr}\widehat{B}_{\tau,U} - \operatorname{tr} B_{\tau,U} = o(w_*)$. The difference $\widehat{B} - D$ is positive semidefinite in fixed dimension, and hence

$$0 \le \|\widehat{B} - D\|_M \le \operatorname{tr}(\widehat{B} - D) = o(w_*).$$

This proves (52). Eigenvalues alone would not identify the operator in these component markings; the polar construction uses the observed averages. Changing an orthonormal basis of a repeated eigenspace leaves the construction unchanged. □

Let $\mathsf{A} = \operatorname{diag}(A_i I_d)$ in orthonormal component matrix coordinates, $d = r^2 - 1$. The ordinary symmetric matrix $\mathsf{Q}_{\tau,U} = \mathsf{A}B_{\tau,U}$ represents the finite energy. Its observed version is $\widehat{\mathsf{Q}} = \mathsf{A}\widehat{B}$. For a simple graph its off-diagonal blocks satisfy

$$-(\mathsf{Q}_{\tau,U})_{ij} = w_e \operatorname{Ad}(U_{ij}) \quad (e = ij). \tag{53}$$

Theorem 9.3. *For the known simple graph and component matrix-algebra markings induced by the unitary frames, the observations above recover projective edge transport consistently whenever w_e/w_* has a fixed positive lower bound. If $\|\widehat{\mathsf{Q}} - \mathsf{Q}\|_{\mathrm{op}} \le \delta w_*$, the recovered adjoint edge map has error at most $C\delta w_*/w_e$ for sufficiently small such ratios of error to weight. In particular all projective edge and cycle transports are recovered uniformly on the common-neck slice, even through reducibility. The cycle representation is recovered up to root conjugation.*

Proof. Divide the observed block in (53) by w_e. Project it, in Frobenius norm, onto the compact embedded group $\operatorname{Ad}(\mathrm{SU}(r))$. Nearest-point projection is unique and smooth in a fixed tubular neighborhood of this compact submanifold of the real matrix space. Norm equivalence in fixed dimension and (52) give the stated error. The marking specifies the matrix algebra and therefore this particular adjoint subgroup. Since $\operatorname{Ad}(\mathrm{SU}(r)) \simeq \mathrm{PSU}(r)$, these are projective transports. An error in a cycle product is bounded by the sum of the edge

errors, because all projected and exact factors are orthogonal. Tree transport then identifies the root representation. Component frame changes act by vertex gauge, giving the stated conjugation ambiguity. □

This is asymptotic recovery with an error bound, not exact reconstruction from one finite-parameter observation. For an edge with $w_e/w_* \to 0$, (52) alone is insufficient after division by w_e. Relative eigenvalue control does not remove this observation-conditioning issue. The areas are fixed known positive numbers; uncertainty or unbounded variation in them is not included.

Central twists are exactly invisible. Multiplying edge maps by elements of the center of $\mathrm{SU}(r)$ leaves the adjoint connection and every marked observation unchanged on each fibre. For fixed projective edge maps the choices of lifts, modulo central vertex changes, form an $H^1(\Gamma; Z(\mathrm{SU}(r)))$ torsor before further stabilizer identifications. Thus an $\mathrm{SU}(r)$ lift cannot be inferred from these data.

9.3. **Compact families with unknown incidence.** Take six labelled copies of $\mathbb{P}^1$, each with three branch points at $0, 1, \infty$. Pair their eighteen branch points according to a connected simple trivalent graph on the six vertices. The observer is given the components and unpaired branch points, but not the pairing. We use the finite class of these pairings, including the triangular prism and $K_{3,3}$.

Proposition 9.4. *Every such pairing has a proper projective holomorphic smoothing with smooth total surface, first-order node equations $uv = t$, and connected genus-four smooth fibres. One can choose smooth ambient Kähler forms so that every normalization component has the same prescribed area-one Fubini–Study metric and every smaller node bidisc has the same metric germ. Component-trivial flat $\mathrm{SU}(r)$ bundles with arbitrary unitary edge maps exist on these entire families.*

Proof. Every central rational component has three distinct nodes, so the curve is stable and its arithmetic genus is $9-6+1=4$. The proper flat analytic plumbing of [HK14, Section 8.3 of the preprint version] uses independent node parameters; restrict to their diagonal $t_e = t$. Concretely, glue product exteriors to disjoint node bidiscs by $(u,t) \mapsto (u, t/u)$ and its reverse. For $|t| < a^2$ the incidence collars at a node are disjoint. Smaller closed exterior cores and node bidiscs cover the fibres over each closed smaller parameter disc, which also verifies properness. The total space is smooth because $uv - t$ has nonzero t derivative at a node, and is a product away from the nodes. Connectedness of the graph gives connected smooth fibres.

The relative dualizing line bundle is invertible and has degree one on each central component. Its cube is very ample and has vanishing first cohomology on every fibre by [DM69, Theorem 1.2]. Riemann–Roch gives fifteen sections. The analytic base-change theorem [Gra60, Section 7, Satz 5] applies to this proper flat family over a reduced disc and the line bundle, which is flat over the base, with constant section dimension. Its direct image is locally free of rank fifteen. After shrinking and trivializing it, evaluation embeds the family in $\mathbb{P}^{14} \times \Delta$. Fibrewise very ampleness separates the nodal tangent directions; the base coordinate supplies the remaining direction. This proves relative projectivity of the analytic plumbing.

An embedding alone would not give identical local metrics for different pairings. To arrange that, use the fixed $\mathcal{O}(1)$ and Fubini–Study metric on each component. Choose unitary Möbius coordinates at the three branch points: z at zero, $(z-1)/(z+1)$ at one, and $1/z$ at infinity. In suitable holomorphic frames their squared metric is $(1+|u|^2)^{-1}$. Pull these line bundles to the product exteriors and use a trivial line on each node bidisc. Identify the chosen branch frames

with its frame on the incidence collars. There are no triple overlaps, so these identifications define a holomorphic line bundle on the entire restricted family.

On a node give its frame squared norm

$$|e|^2 = \big((1+|u|^2)(1+|v|^2)\big)^{-1}.$$

On the u overlap its ratio to the component metric is $(1+|t/u|^2)^{-1}$. Interpolate the logarithm of this ratio on fixed disjoint collars. It and all tangential derivatives restrict to zero on the central branch. The resulting smooth Hermitian metric therefore has Chern form Ω restricting to the prescribed Fubini–Study form on every component, and equal to $\omega_{\mathrm{FS}}(u)+\omega_{\mathrm{FS}}(v)$ on each smaller node bidisc.

This form is positive near every node and vertically positive on the compact remainder of the central fibre. In fixed submersion charts on that remainder its vertical coefficient is bounded below and its other coefficients are bounded. The Schur complement shows that $\Omega + C\pi^*\omega_\Delta$ is positive for a sufficiently large C. The added term is nonnegative near the nodes and vanishes on fibres. Properness allows one smaller base disc on which positivity holds everywhere. Take the largest C over the finite set of pairings. The smaller-node germs and the central component metrics are then identical in their declared coordinates.

Finally the same exterior/node cover with constant special-unitary incidence transitions gives the flat bundles, parallel metric and determinant. This construction verifies all the hypotheses of Proposition 9.1 on these families. □

On either graph one can take tree maps equal to I, two free-cycle maps equal to $(I+i\sigma_x)/\sqrt{2}$ and $(I+i\sigma_y)/\sqrt{2}$, and the other two equal to I. They give irreducible rank-two representations. Thus these same compact families carry stable bundles with genuinely noncommuting gluing and the actual HYM dynamics of the preceding proposition.

9.4. **Incidence and the integral invariant.** For this inverse problem the data are the six labelled unpaired components with their local metrics and area normalization, the rank, the normalized parameter t, the fact that each node is smoothed to first order, and the positive orientation of the base loop. Choose local parallel orthonormal adjoint frames on fixed component exteriors. Observe the first $m = 6d$ eigenvalues and their orthonormal eigensection averages there. The incidence matrix, global vanishing cycles, cycle lattice and monodromy are not part of the data. Arbitrary orthonormal adjoint frames suffice for this topology problem; the stronger matrix-algebra marking is needed only to recover projective bundle transport.

Theorem 9.5. *For the finite family class of Proposition 9.4, the declared observations recover the labelled incidence graph exactly for all sufficiently small $|t|$, uniformly in all unitary transports. They thereby determine the integral symplectic conjugacy class of the positive monodromy $\mathcal{T} : H_1(C_t,\mathbb{Z}) \to H_1(C_t,\mathbb{Z})$. In particular they determine* $\operatorname{Tor}\operatorname{coker}(\mathcal{T}-I)$. *For the triangular prism and $K_{3,3}$ these groups are, respectively,*

$$\mathbb{Z}/5\mathbb{Z}\oplus\mathbb{Z}/15\mathbb{Z}, \qquad (\mathbb{Z}/3\mathbb{Z})^2\oplus\mathbb{Z}/9\mathbb{Z}.$$

Proof. The component areas are one. Let W_t have the measured average columns and put $S_t = W_tW_t^*$. Lemma 9.2 gives invertibility for small t and the consistent estimator

$$\widehat{Q}_t = \frac{1}{2\pi} S_t^{-1/2} W_t \operatorname{diag}(L\lambda_0,\dots,L\lambda_{m-1}) W_t^* S_t^{-1/2}. \tag{54}$$

This formula contains no edge list. For each underlying graph it satisfies $\|\widehat{Q}_t - Q_{\Gamma,U}\|_{\mathrm{op}} \to 0$, where

$$\langle Q_{\Gamma,U} X, X\rangle = \sum_{ij\in E} \|X_i - \mathrm{Ad}(U_{ij})X_j\|^2.$$

The maximum of the finitely many family-dependent errors still tends to zero uniformly in U.

Simplicity of the graph gives $-(Q_{\Gamma,U})_{ij} = \mathrm{Ad}(U_{ij})$ on an edge and zero otherwise. The Frobenius norms are $\sqrt{d}$ and zero. Define

$$ij \in \widehat{E}_t \quad\Longleftrightarrow\quad \| - (\widehat{Q}_t)_{ij}\|_{\mathrm{F}} > \tfrac{1}{2}\sqrt{d}.$$

An operator error less than $1/2$ gives a block Frobenius error less than $\sqrt{d}/2$, proving exact adjacency recovery. Independent orthonormal frame changes preserve these block norms. If the constructed operator error is less than $1/4$, any further operator perturbation of norm less than $1/4$ preserves the conclusion. This is an error margin for the constructed operator; raw measurement-noise propagation through (54) is a separate question.

The recovered object is the labelled adjacency graph. This does not determine which of the three individually marked branch points is attached to each neighbor. The integral monodromy conjugacy class below depends only on the graph and the prescribed smoothing orders, so this remaining ambiguity does not affect the stated inverse conclusion.

Orient the recovered edges and form the integral incidence map $B : \mathbb{Z}^E \to \mathbb{Z}^V$, $B(e) = v_{\mathrm{head}} - v_{\mathrm{tail}}$. Write

$$F = \ker B, \qquad C = \operatorname{im} B^T, \qquad F^* = \mathrm{Hom}(F,\mathbb{Z}) \simeq \mathbb{Z}^E/C.$$

A spanning tree supplies an integral fundamental-cycle basis and proves the required saturation. The cycle lattice F and its dual F^* are not identified isometrically.

For rational normalization components the integral degeneration sequence is

$$0 \longrightarrow F^* \simeq V_{\mathrm{van}} \longrightarrow H_1(C_t,\mathbb{Z}) \longrightarrow F \longrightarrow 0. \tag{55}$$

Here V_{van} is generated by the oriented vanishing loops δ_e, whose relations are $\sum_e B_{ve}\delta_e = 0$. These are the normalization and nearby-cycle descriptions in [AO20, Proposition 7.3.3 and Remark 7.4.2]. They can also be seen by cutting the surface into the rational component pieces and annuli: the boundary relations at each piece are the incidence relations, and the remaining crossing cycles are the graph cycles. The quotient in (55) is free, so V_{van} is primitive. Its disjoint generators make it isotropic; its rank four is half the rank of $H_1(C_t,\mathbb{Z})$.

Use the intersection convention $\langle a_i, b_j\rangle = \delta_{ij}$ and positive local monodromy

$$\mathcal{T}(x) = x - \sum_e \langle x, \delta_e\rangle \delta_e.$$

The coefficient of every node is one because $uv = t$. The integral Picard–Lefschetz factorization [AO20, Theorem 7.4.1, Corollary 7.4.3 and Proposition 7.4.4] says that $N = \mathcal{T} - I$ kills V_{van} and induces

$$q : F \longrightarrow F^*, \qquad q(c)(d) = \sum_e c_e d_e.$$

Choose a fundamental-cycle matrix Z with integral basis columns and set $G_\Gamma = Z^T Z$. Choose the dual basis a of V_{van} and lifts b of the cycle basis. Primitivity permits integral symplectic

lifts: the skew intersection matrix of initial lifts is removed by adding integer combinations of the a_i. In this symplectic basis,

$$\mathcal{T} = \begin{pmatrix} I & G_\Gamma \\ 0 & I \end{pmatrix}, \qquad N = \begin{pmatrix} 0 & G_\Gamma \\ 0 & 0 \end{pmatrix}. \tag{56}$$

The sign agrees with the chosen positive loop and intersection convention.

Changing the cycle basis by $P \in \mathrm{GL}(4,\mathbb{Z})$ replaces G_Γ by $P^T G_\Gamma P$ and conjugates (56) symplectically. Changing the lifts does not alter the induced block. Thus the graph determines the claimed integral symplectic conjugacy class, rather than a matrix in an unprovided homology marking. Reversing the base loop would reverse the sign; that orientation was declared as input.

The positive form G_Γ need not be unimodular. In particular the image of N need not be primitive, although its saturation V_{van} is. We have

$$\operatorname{Tor}\operatorname{coker} N \simeq F^*/q(F) \simeq \operatorname{coker} G_\Gamma.$$

This is the graph critical group. Its order is the spanning-tree number; the cycle-lattice, dual-lattice and critical-group identifications are classical [BHN97, Propositions 1, 3 and 7, and Lemma 2]. The explicit calculation below gives the asserted groups. □

9.5. **Two identical local models.** Number the vertices $0,\dots,5$. The prism has triangles $012, 345$ and matching edges $03, 14, 25$; $K_{3,3}$ has every edge between $\{0,1,2\}$ and $\{3,4,5\}$. Orient every edge from its smaller to its larger vertex and use lexicographic edge order. Choose the prism tree $01, 02, 03, 14, 25$ and the bipartite tree $03, 04, 05, 13, 23$. For each remaining edge, use the oriented chord followed by the returning tree path. These four cycles are an integral basis: their chord submatrix is I_4, and subtracting their chord coefficients from any integral cycle leaves a cycle on a tree, which is zero. Their Gram matrices are

$$G_{\mathrm{pr}} = \begin{pmatrix} 3 & -1 & 1 & 2 \\ -1 & 4 & 1 & -2 \\ 1 & 1 & 4 & 2 \\ 2 & -2 & 2 & 5 \end{pmatrix}, \qquad G_{\mathrm{bi}} = \begin{pmatrix} 4 & 2 & 2 & 1 \\ 2 & 4 & 1 & 2 \\ 2 & 1 & 4 & 2 \\ 1 & 2 & 2 & 4 \end{pmatrix}. \tag{57}$$

Exact integer arithmetic gives

graph	gcds of minors of orders $1,2,3,4$	Smith diagonal	$\det G_\Gamma$
prism	$1, 1, 5, 75$	$1, 1, 5, 15$	75
$K_{3,3}$	$1, 3, 9, 81$	$1, 3, 3, 9$	81

Successive ratios of the minor gcds give the Smith diagonal. The exact calculation also verifies $BZ = 0$, the identity chord submatrix, $N^2 = 0$, and preservation of the standard symplectic form by (56). All matrices and integer checks are reproduced by the accompanying computation.

Both smooth fibres have genus four, and their rational nilpotent monodromy ranks are four. Their smooth surfaces therefore have the same diffeomorphism type. What differs is an integral invariant of the degeneration. The local geometric inputs alone do not determine it; the marked global low-mode data determine the missing adjacency and hence the integral form.

This conclusion uses the full Gram form for the conjugacy class; the Smith diagonal records only its cokernel group. Parallel edges are excluded because sums of their transported off-diagonal blocks can cancel. Unknown smoothing orders would change the monodromy pairing

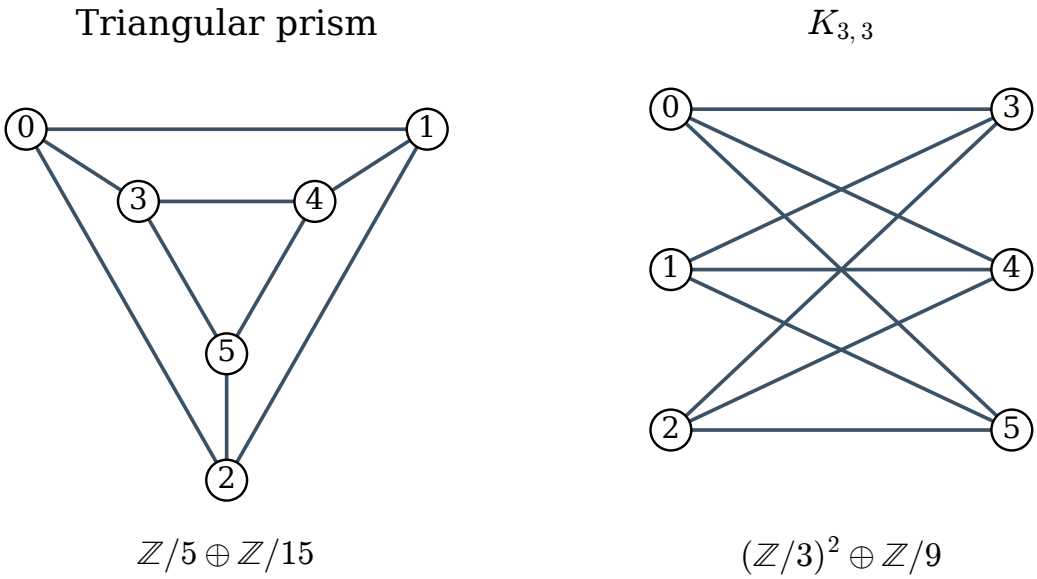


FIGURE 2. Two attachment graphs for the same six triply pointed rational components. The local component metrics and first-order node metric germs can be identical. The integral monodromy torsion groups have different orders, 75 and 81. The diagram depicts the exact attachment data; it is not a numerical spectral experiment.

to $Z^T \operatorname{diag}(n_e) Z$ and require additional observations. Neither the minimality of the present observation package nor recovery from unmarked eigenvalues is asserted.

10. Further questions

The preceding theorems distinguish three kinds of information. The transported energy controls the complete weak linearized spectrum. The nonlinear reconstruction also requires the component and collar estimates for the actual Chern curvature. Integral monodromy is obtained only after the marked observations have determined the missing incidence and the first-order smoothing has fixed its integral pairing. Several extensions require additional arguments at precisely these distinctions.

10.1. **The weakest nonlinear time.** Proposition 3.5 gives a compact stable rank-two family whose weakest eigenvalue has order h^4, whereas its largest edge weight has order h. Thus $L_* \asymp h^{-1}$ and $L_* \nu_0 \asymp h^3$. Theorem 5.2 controls bounded intervals of order h^{-1}. Its residual estimate, integrated to order h^{-4}, gives a bound of order $h^{-2}(1 + |\log h|)^2$, which does not tend to zero. The all-time hypothesis of Theorem 5.3 therefore cannot be removed by the same argument.

Question 10.1. For the family of Proposition 3.5, does a finite nonlinear evolution track the actual HYM flow uniformly through the time h^{-4} for the spatial initial class of Section 6? If so, is the leading equation (19) sufficient, or must it be corrected?

Failure of the integrated upper bound is not a counterexample to tracking. Both exact and finite solutions can retain a weak component for a long time. A useful proof might estimate

the residual relative to the evolving dissipation, or incorporate the next terms in the finite field. Either approach must retain cancellations among the incident forces; a bound on each force separately can lose the scale of their sum. Relative linear spectral comparison alone does not settle this question.

10.2. **Independent nodes with component holonomy.** Section 7 permits arbitrary ratios of the component energy to the common neck energy. Sections 2–6 permit arbitrary ratios of the six node energies when the component connections are trivial. Combining these results requires a compact family with both sets of independent parameters and an analysis on that family. In particular, the harmonic gauges on the punctured components must give energy-controlled boundary estimates at each individual neck, and the nonlinear matching must retain the component connection terms. An abstract sum of the two proved operators would not verify these requirements.

10.3. **Nonunitary degeneration and singular limits.** All bundles used in this paper admit flat unitary reference connections. Their holomorphic structures are polystable, and the degenerations of stability studied here concern reducibility and the appearance of parallel endomorphisms. Genuine nonunitary extensions can introduce additional deformation and obstruction terms into the effective equation. Unstable holomorphic bundles also lack the stationary reference metric used in the order bounds and tail estimates. Their actual HYM evolution requires a different normalization and confinement argument.

On a smooth curve, varying the Kähler form alone does not produce ordinary slope walls. A broader stability problem must be realized by varying holomorphic bundle data, a specified parabolic stability structure, or higher-dimensional polarizations. When singular graded sheaves or curvature concentration occur, their local charge and scale data must enter the matching problem. No singular concentration law follows from the flat-bundle results proved here.

10.4. **Conditioning and additional observations.** For a known graph with independent weights, Theorem 9.3 resolves an edge only when the observed block error is small relative to that edge's weight. An observation scheme adapted to successively weaker spectral subspaces might recover smaller weights, but would require compatible markings and a sharper operator estimate. Eigenvalue accuracy by itself does not provide these identifications.

For unknown incidence, Theorem 9.5 uses a finite simple-graph class, known first-order node smoothing and component averages of eigenvectors. Parallel edges, unknown smoothing multiplicities and unmarked component permutations require different observations. It would be useful to identify a minimal sufficient observation package and to quantify the propagation of errors from raw eigenvalue and eigensection measurements. The present operator-level error margin does not answer that statistical or numerical question.

10.5. **Higher-dimensional geometry.** The curve construction separates component parallel endomorphisms from the directions coupled by the nodes. In higher dimensions both that decomposition and the local capacity law can change. A bundle theorem must identify the actual endomorphism complex, construct its compact holomorphic family, and prove the uniform analytic estimates in the chosen Kähler metrics. Small eigenvalues of ordinary differential forms do not automatically give the required endomorphism modes. Likewise, recovering projective bundle transport does not recover the integral cohomology of the base. A higher-dimensional synthesis must specify and prove each connection on the same geometric family.

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